# Schwarz Coupling for Solid Mechanics in Norma: Formulation, Algorithms, and Verification

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**Abstract**

This note specifies the Schwarz alternating method as implemented in Norma for the coupling of finite element subdomains in solid mechanics, in the form needed to implement the same methods in another code. A multidomain simulation advances in controller steps, called stops; within a stop each subdomain is advanced by its own time integrator, with its own mesh and time step, and the subdomains are solved in turn with interface data exchanged between the solves until the data converge. The note states the controller (stops, substeps, the time interpolation of partner data, the stopping criteria, the relaxation and acceleration of the iteration, and the control of the Newton tolerance of the subdomain solves), the interface transfer operators (interpolation in a partner mesh and $L^2$ projections built from boundary mass matrices), and the five transmission conditions in the code: the overlapping Dirichlet coupling, the classical Dirichlet–Neumann exchange on a shared interface, the matching Dirichlet–Neumann exchange, the Robin–Robin condition, and the contact coupling. The matching exchange is treated in the most detail, because it is the only one whose fixed point conserves the energy of a dynamic problem: its Dirichlet side matches one kinematic quantity of the partner, the velocity or the displacement, and derives the other two from its own Newmark relations; its Neumann side receives the d'Alembert reaction of the partner through the transpose of the kinematic transfer, both transfers built from one cross-mass matrix; the iteration stops on the interface jump and the force residual; and the initial acceleration is computed on the coupled system. At the fixed point the exchange solves the velocity-constrained coupled problem of Gravouil and Combescure, of Prakash and Hjelmstad, and of Connors, Owen, Kuberry, and Bochev, for which the rate invariant (the energy functional evaluated on the velocity and the acceleration, the invariant of the time-differentiated system) is conserved exactly for linear elasticity, for any pair of Newmark schemes, for any mesh ratio, and, with the coarse-step side as the Dirichlet side, for any integer step ratio. The note gives the formulation and its proof, the algorithm, a direct interface solve for two explicit subdomains, and the verification on a cantilever beam and on nested cylinders: the matched energy is constant to the Schwarz tolerance over $10^4$ steps in every configuration measured, and to roundoff with the direct solve, while the classical exchange inverts an element within 1 ms with implicit subdomains. An appendix lists the input keys of every coupling and the tests and examples that exercise them.

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# 1 Scope

A solid body $\Omega \subset \mathbb{R}^3$ is decomposed into subdomains $\Omega_k$ that are discretized and integrated in time independently, each with its own mesh, its own time integrator, and its own time step, and coupled by a Schwarz iteration: within each time window the subdomains are solved in turn, exchanging interface data, until the data converge, following Mota, Tezaur, and Phlipot [1]. Norma implements five transmission conditions, each a boundary condition type of a subdomain input:

1. the *overlapping Dirichlet coupling* (`Schwarz overlap`): the subdomains overlap, and each receives on its boundary inside the partner the displacement, velocity, and acceleration interpolated from the partner's solution (Sec. 6);
2. the *classical Dirichlet–Neumann exchange* (`Schwarz DN nonoverlap` without `matching`): the subdomains share an interface; one side receives the three kinematic fields of the other and the other receives the internal force of the first (Sec. 7);
3. the *matching Dirichlet–Neumann exchange* (`Schwarz DN nonoverlap` with `matching`): one side matches the velocity or the displacement of the other and the other receives the d'Alembert reaction of the first, so that the fixed point of the iteration is the constrained coupled problem whose rate invariant is conserved (Sec. 8);
4. the *Robin–Robin condition* (`Schwarz RR nonoverlap`): each side imposes $\boldsymbol{t} + \alpha \boldsymbol{W} \boldsymbol{u} = \boldsymbol{g}$ on the shared interface, with the datum $\boldsymbol{g}$ formed from the partner's traction and displacement (Sec. 9);
5. the *contact coupling* (`Schwarz contact`): two bodies whose surfaces may come into contact are coupled by a Dirichlet–Neumann exchange that is active while contact is detected (Sec. 10).

All five share the controller of Sec. 3, the relaxation of Sec. 4, and the transfer operators of Sec. 5. The note describes the code as of October 2026 and states, for each method, what it requires of the input and what it does not do.

The requirements on a dynamic coupling, which only the matching exchange satisfies in full, are:

1. *Energy conservation.* At the converged state the coupling does no work on the system: the coupled problem conserves the quantity that the subdomain integrators conserve, to the tolerance of the Schwarz iteration.
2. *Nonconforming meshes.* The meshes need not share nodes on the interface.
3. *Heterogeneous integrators and time steps.* Each subdomain may use an implicit Newmark scheme or central difference, and the time steps may differ by an integer ratio.
4. *Partitioned solution.* Each subdomain is advanced by its own solver; the coupling requires of a subdomain only operations that a structural dynamics code already provides (Sec. 8.9).

The classical exchange satisfies the last three but not the first: with implicit subdomains on the beam of Sec. 11 it inverts an element within 1 ms, and with explicit subdomains it conserves the energy only to 2 to 4% (Sec. 11.8). The overlapping coupling exchanges data on two different surfaces, so there is no shared interface on which the works of the two sides can cancel. The Robin–Robin condition is not absorbing and is used for quasi-static problems.

## 2 Setting and notation

**Semi-discrete equations.** After finite element discretization in space, subdomain $k$ satisfies

$$\boldsymbol{M}_k\,\boldsymbol{a}_k + \boldsymbol{f}_k^{\text{int}}(\boldsymbol{u}_k) = \boldsymbol{f}_k^{\text{ext}} + \boldsymbol{C}_k^T\boldsymbol{\lambda}_k, \tag{1}$$

with $\boldsymbol{u}_k$, $\boldsymbol{v}_k$, $\boldsymbol{a}_k$ the nodal displacement, velocity, and acceleration, $\boldsymbol{M}_k$ the mass matrix (lumped for central difference), $\boldsymbol{f}_k^{\text{int}}$ the internal force, $\boldsymbol{f}_k^{\text{ext}}$ the external force, $\boldsymbol{C}_k$ the Boolean restriction of the nodal vector to the interface nodes, and $\boldsymbol{\lambda}_k$ the interface force acting on subdomain $k$. For linear elasticity $\boldsymbol{f}_k^{\text{int}} = \boldsymbol{K}_k\boldsymbol{u}_k$. A quasi-static subdomain omits the inertial term. The reference coordinates of the nodes are $\boldsymbol{X}$, and the current coordinates are $\boldsymbol{x} = \boldsymbol{X} + \boldsymbol{u}$.

**Time integrators.** A dynamic subdomain is advanced by the Newmark scheme with parameters $(\beta, \gamma)$ and step $\Delta t$:

$$\boldsymbol{u}^{n+1} = \boldsymbol{u}^n + \Delta t\,\boldsymbol{v}^n + \Delta t^2\big[(\tfrac{1}{2} - \beta)\,\boldsymbol{a}^n + \beta\,\boldsymbol{a}^{n+1}\big], \qquad \boldsymbol{v}^{n+1} = \boldsymbol{v}^n + \Delta t\big[(1-\gamma)\,\boldsymbol{a}^n + \gamma\,\boldsymbol{a}^{n+1}\big]. \tag{2}$$

The trapezoidal rule ($\beta = 1/4$, $\gamma = 1/2$, implicit, denoted I) and central difference ($\beta = 0$, $\gamma = 1/2$, explicit with a lumped mass, denoted E) are the two schemes used in the verification. With the predictors $\boldsymbol{u}^{\text{pred}} = \boldsymbol{u}^n + \Delta t\,\boldsymbol{v}^n + (\frac{1}{2} - \beta)\Delta t^2\boldsymbol{a}^n$ and $\boldsymbol{v}^{\text{pred}} = \boldsymbol{v}^n + (1-\gamma)\Delta t\,\boldsymbol{a}^n$, the scheme gives

$$\boldsymbol{a}^{n+1} = \frac{\boldsymbol{v}^{n+1} - \boldsymbol{v}^{\text{pred}}}{\gamma\,\Delta t}, \qquad \boldsymbol{u}^{n+1} = \boldsymbol{u}^{\text{pred}} + \beta\,\Delta t^2\,\boldsymbol{a}^{n+1}, \tag{3}$$

so a prescribed end-of-step velocity determines the acceleration and the displacement of the step, and, for $\beta > 0$,

$$\boldsymbol{a}^{n+1} = \frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^{\text{pred}}}{\beta\,\Delta t^2}, \qquad \boldsymbol{v}^{n+1} = \boldsymbol{v}^{\text{pred}} + \gamma\,\Delta t\,\boldsymbol{a}^{n+1}, \tag{4}$$

so a prescribed end-of-step displacement determines the other two. An implicit subdomain solves its step by Newton's method on the displacement with the Hessian minimizer of Norma; an explicit subdomain divides the residual by the lumped mass. The Newmark predictor and corrector update only the free degrees of freedom of a subdomain, so a value prescribed on a degree of freedom before the step persists through it.

**Prescribed rows.** A boundary condition prescribes a value on a set of degrees of freedom and marks them not free. Before every subdomain solve the boundary force vector is cleared, every degree of freedom is marked free, and the boundary conditions are applied in order, those whose type name ends in `Dirichlet` last, so that a prescribed displacement overrides a coupling datum on a shared node. The coupling conditions of this note either prescribe interface degrees of freedom (Dirichlet type) or add to the boundary force vector (Neumann type). The Newmark initial acceleration solves $\boldsymbol{M}_{ff}\,\boldsymbol{a}_f = (\boldsymbol{f}^{\text{ext}} - \boldsymbol{f}^{\text{int}})_f - \boldsymbol{M}_{fc}\,\boldsymbol{a}_c$ for the free rows $f$ given the prescribed rows $c$. The rows held by a Schwarz coupling carry the partner's projected acceleration, which is zero until the partner has computed its own. On a start from initial conditions those rows are therefore excluded from the set $c$ of the correction; they are included only in the refinement passes of Sec. 3.6 and in the coupled initial acceleration of Sec. 8.6, once every subdomain has an acceleration.

**Roles.** In a Dirichlet–Neumann exchange the *Dirichlet side D* receives a kinematic datum on its interface nodes and the *Neumann side N* receives a force. A pair of integrators is written with the integrator of one named subdomain first (II, IE, EI, EE).

**Interface operators.** Let $\varphi_m^D$ and $\varphi_n^N$ be the interface traces of the nodal basis functions of the two sides of a shared interface $\Gamma$. The cross-mass matrix and the boundary mass matrices are

$$B_{mn} = \int_\Gamma \varphi_m^D \, \varphi_n^N \, \mathrm{d}S, \qquad (\boldsymbol{W}_D)_{ml} = \int_\Gamma \varphi_m^D \, \varphi_l^D \, \mathrm{d}S, \qquad (\boldsymbol{W}_N)_{nl} = \int_\Gamma \varphi_n^N \, \varphi_l^N \, \mathrm{d}S, \tag{5}$$

assembled once on the reference configuration, per Cartesian component. The kinematic transfer to the Dirichlet side and its transpose are

$$\Pi_D = \boldsymbol{W}_D^{-1} \boldsymbol{B}, \qquad \Pi_D^T = \boldsymbol{B}^T \boldsymbol{W}_D^{-1}. \tag{6}$$

$\Pi_D$ maps the interface values of side $N$ to the $L^2$ projection of their trace onto the trace space of side $D$; it reproduces constants when $\boldsymbol{B}\,\mathbf{1} = \boldsymbol{W}_D \mathbf{1}$, that is, when the quadrature of $\boldsymbol{B}$ is exact on products of the two trace spaces. Section 5 gives the assembly of these matrices and the variants that the other couplings use.

**Stops and windows.** The Schwarz controller advances in *stops* of length $\Delta t_c$. Within a stop each subdomain advances to the stop end in its own steps; the *Schwarz iteration* repeats a solve of each subdomain with interface data exchanged between the solves until the data converge, and the stop is then committed. A stop that spans $m$ steps of a subdomain is a *window* of $m$ substeps for that subdomain.

**Energies.** The energy is

$$E = \sum_k \Big( \tfrac{1}{2} \, \boldsymbol{v}_k^T \boldsymbol{M}_k \, \boldsymbol{v}_k + \mathrm{SE}(\boldsymbol{u}_k) \Big), \tag{7}$$

with SE the strain energy. For central difference the kinetic term is evaluated in the staggered form

$$\tfrac{1}{2} \, (\boldsymbol{v}^{n-1/2})^T \boldsymbol{M} \, \boldsymbol{v}^{n+1/2} = \tfrac{1}{2} \, (\boldsymbol{v}^n)^T \boldsymbol{M} \, \boldsymbol{v}^n - \tfrac{\Delta t^2}{8} \, (\boldsymbol{a}^n)^T \boldsymbol{M} \, \boldsymbol{a}^n, \tag{8}$$

the product of the two half-step velocities that bracket $t_n$, written in terms of the integer-step velocity $\boldsymbol{v}^n = \frac{1}{2}(\boldsymbol{v}^{n-1/2} + \boldsymbol{v}^{n+1/2})$ and the acceleration, since $\boldsymbol{v}^{n\pm1/2} = \boldsymbol{v}^n \pm \frac{\Delta t}{2}\boldsymbol{a}^n$. With the strain energy at $\boldsymbol{u}^n$ it is the quadratic $\frac{1}{2}(\boldsymbol{v}^{n-1/2})^T \boldsymbol{M} \boldsymbol{v}^{n+1/2} + \frac{1}{2}(\boldsymbol{u}^n)^T \boldsymbol{K} \boldsymbol{u}^n$ that central difference conserves exactly on a linear problem, evaluated at the integer step; it is not the kinetic energy at a half step, which is $\frac{1}{2}(\boldsymbol{v}^{n+1/2})^T \boldsymbol{M} \boldsymbol{v}^{n+1/2} = \frac{1}{2}(\boldsymbol{v}^n)^T \boldsymbol{M} \boldsymbol{v}^n + \frac{\Delta t}{2}(\boldsymbol{v}^n)^T \boldsymbol{M} \boldsymbol{a}^n + \frac{\Delta t^2}{8}(\boldsymbol{a}^n)^T \boldsymbol{M} \boldsymbol{a}^n$ and is conserved only together with the product strain energy $\frac{1}{2}(\boldsymbol{u}^n)^T \boldsymbol{K} \boldsymbol{u}^{n+1}$. The form (8) pairs with $\mathrm{SE}(\boldsymbol{u}^n)$ and therefore extends to nonlinear materials. The consistent-mass kinetic energy $\frac{1}{2}(\boldsymbol{v}^n)^T \boldsymbol{M} \boldsymbol{v}^n$ at integer steps misrepresents the short-wavelength content of a sharp front by several percent. For linear elasticity the rate invariant is

$$\tilde{E} = \sum_k \Big( \tfrac{1}{2} \, \boldsymbol{a}_k^T \boldsymbol{S}_k \, \boldsymbol{a}_k + \tfrac{1}{2} \, \boldsymbol{v}_k^T \boldsymbol{K}_k \, \boldsymbol{v}_k \Big), \qquad \boldsymbol{S}_k = \boldsymbol{M}_k + \Big( \beta_k - \frac{\gamma_k}{2} \Big) \Delta t_k^2 \, \boldsymbol{K}_k, \tag{9}$$

the energy functional evaluated on the velocity and the acceleration. The time derivative of a solution of $\boldsymbol{M}\ddot{\boldsymbol{u}} + \boldsymbol{K}\boldsymbol{u} = \boldsymbol{f}$ solves $\boldsymbol{M}\dot{\boldsymbol{a}} + \boldsymbol{K}\boldsymbol{v} = \dot{\boldsymbol{f}}$, so this quadratic is the invariant of the time-differentiated system; it is not an energy, and its units are those of energy divided by time squared. $\boldsymbol{S}_k$ is $\boldsymbol{M}_k$ for the trapezoidal rule and $\boldsymbol{M}_k - \frac{1}{4}\Delta t_k^2 \boldsymbol{K}_k$ for central difference, positive definite exactly below the stability limit of central difference. Relative to $E$, $\tilde{E}$ weights each vibration mode by the square of its frequency.

# 3 The controller and the Schwarz iteration

A multidomain input (`type: multi`) lists the subdomain inputs in `domains` and sets the time window, the controller step, and the parameters of the iteration (Appendix A). The controller supplies the time window to every subdomain and clamps each subdomain's step to the controller step.

## 3.1 Stops and substeps

The controller time after stop $s$ is $t_s = t_0 + s\,\Delta t_c$; the number of stops is $\mathrm{round}((t_f - t_0)/\Delta t_c)$, and the span $t_f - t_0$ must be a multiple of $\Delta t_c$. The step of subdomain $k$ is $\Delta t_k = \min(\Delta t_k^{\mathrm{in}}, \Delta t_c)$, with $\Delta t_k^{\mathrm{in}}$ its own input step (default $\Delta t_c$); for central difference it is further limited to the stable step of the mesh. Within a stop the subdomain takes substeps of $\Delta t_k$ from $t_{s-1}$, the last substep taking a remainder of up to $1.5\,\Delta t_k$, until it reaches $t_s$ to a relative tolerance of $10^{-6}$. A stop of $m_k$ substeps is a window of $m_k$ for subdomain $k$. No check enforces that $\Delta t_c$ is a multiple of $\Delta t_k$; the substep count is whatever this rule produces.

## 3.2 One Schwarz iteration

At the start of a stop the controller saves the state of every subdomain at $t_{s-1}$ (displacement, velocity, acceleration, internal force, and the material state), clears the partner histories and the relaxation state (Sec. 4), swaps the roles of the pairs that request it (Sec. 7), and, if the interface predictor is enabled, forms the predictor of Sec. 7.1. Iteration $i$ of the stop then proceeds as follows.

1. Every subdomain is reset to $t_{s-1}$.
2. The subdomains are advanced one after another, in the order of `domains`. Subdomain $k$ first clears its own history and records the anchor $(t_{s-1}, \boldsymbol{u}_k, \boldsymbol{v}_k, \boldsymbol{a}_k, \boldsymbol{f}_k^{\mathrm{int}})$; then for each substep it applies its boundary conditions at the substep end time, solves the substep, and records $(t, \boldsymbol{u}_k, \boldsymbol{v}_k, \boldsymbol{a}_k, \boldsymbol{f}_k^{\mathrm{int}})$.
3. A coupling condition of subdomain $k$ applied at time $t$ reads the partner $j$'s displacement, velocity, acceleration, and internal force by piecewise-linear interpolation in time of the partner's history,
$$\boldsymbol{x}_j(t) = \frac{t_{l+1} - t}{t_{l+1} - t_l}\,\boldsymbol{x}_j^{\,l} + \frac{t - t_l}{t_{l+1} - t_l}\,\boldsymbol{x}_j^{\,l+1}, \qquad t_l \le t \le t_{l+1}, \tag{10}$$
with the end values returned outside the recorded range. If the partner's history is empty, the condition uses the predictor when one is enabled, the partner's state at $t_{s-1}$ otherwise.
4. After the pass over all subdomains the convergence tests of Sec. 3.3 are evaluated. If the stop has not converged, the end-of-iteration state is saved as the comparison iterate for the next test, and step 1 is repeated.

The iteration is multiplicative: subdomain $k$ reads from a partner listed before it the history of the current iteration, and from a partner listed after it the history of the previous iteration; at $i = 0$ the latter is empty, so the first-listed subdomain reads its partner's state at $t_{s-1}$, constant in time, or the predictor. Every iteration restarts the subdomain solves from the state at $t_{s-1}$, not from the previous iterate.

### 3.3 Stopping criteria

**Displacement update.** With $\Delta t_c$ the controller step and $\boldsymbol{w}_k = \boldsymbol{u}_k + \Delta t_c\, \boldsymbol{v}_k$ over all degrees of freedom of subdomain $k$ (prescribed ones included), the update between iterations $i-1$ and $i$ is

$$\Delta U = \Big(\sum_k \|\boldsymbol{w}_k^{i} - \boldsymbol{w}_k^{i-1}\|^2\Big)^{1/2}, \qquad \Delta u = \Delta U \Big/ \Big(\sum_k \|\boldsymbol{w}_k^{i}\|^2\Big)^{1/2}, \tag{11}$$

and the stop has converged when $\Delta U \le$ `absolute tolerance` or $\Delta u \le$ `relative tolerance`. At $i = 0$ the comparison iterate is the state at $t_{s-1}$, so $\Delta U$ is the change over the whole step and the relative test does not apply. This criterion stops the overlapping, classical Dirichlet–Neumann, Robin–Robin, and contact couplings.

**Matching criterion.** A matching pair converges when the one-sided jump of its matched quantity and its force residual are at or below `matching relative tolerance`, as defined in Sec. 8.5. When every Schwarz coupling of the simulation is a matching pair, this criterion alone decides, also at $i = 0$, and the controller's `relative tolerance` and `absolute tolerance` are not used (given, they are ignored with a warning). When the simulation has couplings of both kinds, both criteria must hold.

**Iteration count.** A stop ends at iteration $i \ge 1$ when the criterion holds and $i \ge$ `minimum iterations`, or when $i >$ `maximum iterations`, or when $\Delta U = 0$. At $i = 0$ it ends without further iterations when $\Delta U \le$ `absolute tolerance` (or, for a simulation of matching pairs only, when the matching criterion holds) and every matching pair has converged. A stop that reaches `maximum iterations` without converging aborts the run (`unconverged step action: abort`, the default) with the measures that apply, or continues with the last iterate (`warn`).

**Iterations not accepted.** An iteration at which a relaxation factor of magnitude at most $10^{-12}$ was applied is not accepted as converged, because a factor of zero leaves the interface datum unchanged and the subdomains then return the solution they already had (Sec. 4.1). Under `tight final solve` (Sec. 3.5) convergence is accepted only at an iteration whose Newton tolerance was the floor.

### 3.4 Windowed stops and relaxation slots

The relaxation state of a coupling pair (Sec. 4) is kept per pair and per *slot*, a substep end time at which the relaxed side applies its condition; a stop that spans several substeps of the relaxed side has several slots. The state is cleared at every stop. A state shared across substeps would combine iterates of different times, which acts as a filter on the exchanged datum and moves the converged solution. Aitken and Anderson acceleration apply to stops of one slot; a stop of several slots uses the fixed factor `relaxation parameter`. Windowed and single-substep runs of the same problem agree to the Schwarz truncation error, and the total number of substep solves is proportional to the iterations per stop, so a window does not reduce the number of subdomain solves.

### 3.5 Control of the Newton tolerance of the subdomain solves

The optional block `inexact Newton` sets the relative Newton tolerance of every implicit subdomain before the solves of each Schwarz iteration: the ceiling `maximum tolerance` (default $10^{-2}$) at $i = 0$, and

$$\tau_i = \mathrm{clamp}\big(\eta\, r_{i-1},\ \tau_{\min},\ \tau_{\max}\big), \qquad i \ge 1, \tag{12}$$

with $\eta$ the `tolerance ratio` (default 0.01), $\tau_{\min}$ the `minimum tolerance` (default $10^{-12}$), and $r_{i-1}$ the relative Schwarz residual of the previous iteration: the matching measure of Sec. 8.5 when every coupling is a matching pair, $\Delta u$ of (11) when there is none, and the larger of the two otherwise. A Newton solve stops when its residual norm is at or below the solver's absolute tolerance or its ratio to the initial residual norm is at or below $\tau_i$; the absolute tolerance of the subdomain `solver` block still applies. Because every Schwarz iteration restarts from the state at $t_{s-1}$, a constant Newton tolerance larger than the Schwarz tolerance leaves an error of the size of the Newton error that the displacement criterion does not detect; the ratio $\eta < 1$ keeps the Newton error of the accepted iterate below the Schwarz update by the factor $\eta$ (Sec. 11.7). With `tight final solve: true` a convergence reached at $\tau_i > \tau_{\min}$ is rejected and the iteration continues with $\tau_{\min}$ for the rest of the stop. The control acts on Hessian minimizer solvers only; explicit subdomains are not affected.

### 3.6 Initialization

Before the first stop the controller builds the transfer operators of every coupling (Sec. 5) on the reference configuration, applies the initial conditions, records a history snapshot of every subdomain, applies the boundary conditions, and computes the initial acceleration of every dynamic subdomain with the coupling rows excluded from the correction of Sec. 2. If the run is a restart, or a Robin–Robin interface has nonzero initial displacement or velocity, two refinement passes follow, each applying the boundary conditions from the updated histories and recomputing the initial accelerations with the coupling rows included in the correction of Sec. 2. Matching pairs then compute the coupled initial acceleration of Sec. 8.6, and contact pairs evaluate their detection (Sec. 10).

### 3.7 Output

Exodus and CSV output are written at the controller stops that are multiples of `Exodus output interval` (default $\Delta t_c$) and `CSV output interval` (default off). With `blended energy output: true` the controller writes `<name>-energy.csv` at every stop, with the columns `time`, `stored_energy`, `kinetic_energy`, `total_energy` ($E$ of (7), blended over overlaps as in Sec. 6.2), `rate_invariant_total` and one `rate_invariant_<domain>` per subdomain ($\tilde{E}$ of (9), NaN unless every material is linear elastic and every integrator is central difference or Newmark without HHT-$\alpha$), `staggered_kinetic_interface` and `staggered_kinetic_interior` (the central difference kinetic term split between interface and interior nodes), and, for every Dirichlet–Neumann pair, the relative one-sided jumps `displacement_jump_<D>_<N>` and `velocity_jump_<D>_<N>` of Sec. 8.5 with their root mean square values (`_rms`) and, for matching pairs, the impulse residual (29) per component and its relative value. The rate invariant and staggered columns are not blended over overlaps.

### 3.8 Swaps and restart

A `swaps` list replaces a subdomain by another input between stops when a criterion holds: a time, a stress recovery estimate, an elastic to plastic transition, or the overlap $L^2$ error of Sec. 6.1 against a tolerance in a direction (`refine` or `coarsen`). The replacement receives the state of the subdomain it replaces, by copy on the same mesh or by $L^2$ transfer between meshes; the coupling conditions and transfer operators are rebuilt after a swap. A restart of a multidomain run resumes every subdomain from the same checkpoint time; it is allowed with the overlapping coupling, warned against with the nonoverlapping couplings, and rejected with contact.

# 4 Relaxation and acceleration of the iteration

The datum that a coupling transmits is relaxed on one side of each pair: the Dirichlet side of a Dirichlet–Neumann or contact pair, and the later-listed side of a Robin–Robin pair. The overlapping coupling is not relaxed. Let $\boldsymbol{g}_i$ be the datum computed from the partner at iteration $i$ and $\boldsymbol{\lambda}_{i-1}$ the datum applied at the previous iteration of the same slot; a new slot has no previous datum and uses $\boldsymbol{g}_i$ unrelaxed (the Robin–Robin datum starts from zero, Sec. 9). The input key `relaxation` selects one of four schemes, with `relaxation parameter` $\theta_0$ (default 1; 0.5 under Anderson acceleration).

**Fixed.** $\boldsymbol{\lambda}_i = \theta_0\, \boldsymbol{g}_i + (1-\theta_0)\, \boldsymbol{\lambda}_{i-1}$.

## 4.1 Aitken relaxation

With the residual $\boldsymbol{r}_i = \boldsymbol{g}_i - \boldsymbol{\lambda}_{i-1}$ and $\boldsymbol{\delta}_i = \boldsymbol{r}_i - \boldsymbol{r}_{i-1}$, the *recursive* form (`aitken recursive`, Irons and Tuck) applies $\boldsymbol{\lambda}_i = \theta_i \boldsymbol{g}_i + (1-\theta_i)\boldsymbol{\lambda}_{i-1}$ with

$$\theta_i = -\,\theta_{i-1}\,\frac{\boldsymbol{r}_{i-1}^T \boldsymbol{\delta}_i}{\|\boldsymbol{\delta}_i\|^2}, \tag{13}$$

and the *secant* form (`aitken secant`) applies

$$\theta_i = -\,\frac{\boldsymbol{d}_i^T \boldsymbol{\delta}_i}{\|\boldsymbol{\delta}_i\|^2}, \qquad \boldsymbol{d}_i = \boldsymbol{\lambda}_{i-1} - \boldsymbol{\lambda}_{i-2}. \tag{14}$$

Both use $\theta_0$ for the first `aitken NO parameter` iterations (default 1) and whenever the required previous residuals do not exist. The factor is not clamped. The update is skipped, and the previous factor (recursive) or $\theta_0$ (secant) kept, when $\|\boldsymbol{\delta}_i\|^2 \le (10^{-14})^2 \|\boldsymbol{\lambda}_{i-1}\|^2$, so that a pair whose datum has stopped moving does not produce a factor of zero. A factor of magnitude at most $10^{-12}$ is recorded, and the iteration at which it was applied is not accepted as converged (Sec. 3.3): a zero factor leaves the interface datum at its previous value, every subdomain then re-solves against the data it already had, and the displacement update that follows measures the effect of an unchanged datum, not the interface residual. For a Dirichlet–Neumann pair the first secant factor appears at iteration 2: at iteration 0 the new slot has $\boldsymbol{r} = 0$, and at iteration 1 the difference $\boldsymbol{d}$ is zero.

The vector on which the factor is formed depends on the coupling. For the classical Dirichlet–Neumann and contact exchanges it is the partner's whole displacement field; for the matching exchange it is the projected interface trace $\Pi_D \boldsymbol{q}_N$ of the matched quantity, because the interior values do not enter the exchange, follow a map of zero gain, and dominate the residual, which drives the factor toward 1 where the interface map, of gain near $-1$, does not contract. For the Robin–Robin condition it is the Robin datum itself.

## 4.2 Anderson acceleration

Anderson acceleration [4] (`anderson`), in the interface quasi-Newton form of Degroote, Bathe, and Vierendeels [5], combines the last $m$ iterates (`anderson depth`, default 10). With $\boldsymbol{x}_i$ the datum applied at iteration $i$, $\boldsymbol{f}_i = \boldsymbol{g}_i - \boldsymbol{x}_i$ its residual, $\boldsymbol{t}_i$ the trace of the residual on which the coefficients are formed, and $\Delta\boldsymbol{X}$, $\Delta\boldsymbol{F}$, $\Delta\boldsymbol{T}$ the matrices of differences of the last $m+1$ iterates,

$$\boldsymbol{\gamma} = \arg\min \|\boldsymbol{t}_i - \Delta\boldsymbol{T}\,\boldsymbol{\gamma}\|, \qquad \boldsymbol{x}_{i+1} = \boldsymbol{x}_i + \beta\,\boldsymbol{f}_i - (\Delta\boldsymbol{X} + \beta\,\Delta\boldsymbol{F})\,\boldsymbol{\gamma}, \tag{15}$$

with the mixing parameter $\beta$ (`relaxation parameter`, default 0.5). The least-squares problem is solved by QR, with the oldest column dropped while the condition number of the triangular factor

exceeds $10^{10}$. The first iteration of a slot has no history and is the fixed relaxation $\boldsymbol{x}_1 = \boldsymbol{x}_0 + \beta \boldsymbol{f}_0$; it is not entered in the history. With $\beta = 1$ and a depth at least the iteration count the iterate equals the fixed-point map applied to the GMRES iterate of the same index on a linear interface map (Walker and Ni, Theorem 2.2), which test 136 checks to $10^{-8}$ on a probed interface map. The trace $\boldsymbol{t}_i$ is the projected interface residual $\Pi_D(\boldsymbol{g}_i - \boldsymbol{x}_i)$ of the displacement for the classical and contact exchanges (one set of coefficients then updates the displacement, velocity, and acceleration, each with its own history), of the matched quantity for the matching exchange, and the residual of the Robin datum for the Robin–Robin condition. The history of a pair is cleared at every stop and after a stall (Sec. 8.5). On the beam of Sec. 11 (1 ms, $\epsilon = 10^{-12}$) $\beta = 0.5$ needs 7.0, 23.1, and 7.9 iterations per window on the implicit pair, on the mixed pair with the explicit side as the Dirichlet side, and on the explicit pair at mesh ratio 1:0.5 with 4:1 steps, against 8.9, 28.7, and 8.8 at $\beta = 1$, with the same energies; for the matching exchange it is the recommended scheme.

# 5 Interface transfer operators

## 5.1 Point location in a partner mesh

A node or quadrature point of one subdomain is located in the volume mesh of a partner by a search over the elements of a named block (`source block`) with a bounding-box grid built on the reference configuration; a candidate element is accepted when the point lies within its bounding box enlarged by 10% and its parametric coordinates $\boldsymbol{\xi}$, found by Newton's method on $\sum_a N_a(\boldsymbol{\xi}) \boldsymbol{X}_a = \boldsymbol{X}$ to $10^{-8}$, satisfy the parametric inside test enlarged by the `search tolerance` (default $10^{-6}$, dimensionless): for a hexahedron $|\xi_c| \le 1+\text{tol}$ in every direction, for a tetrahedron $\sum_c \xi_c \le 1+\text{tol}$. The first element that passes is used, and a point found in no element aborts the run. The shape function values $N_a(\boldsymbol{\xi})$ are stored once, on the reference configuration, and reused at every step.

A point of a facet of one side of a shared interface is located on the partner's side set by a closest-point projection onto the partner facets, with a bounding-box prefilter by the nearest partner node; the facet whose parametric overshoot is at most $10^{-6}$ is preferred, with the distance deciding between equal overshoots, and the facet of the nearest node is used when no facet satisfies the overshoot bound. The search uses the partner's current coordinates, which are the reference coordinates when the operators are built at initialization.

## 5.2 Boundary mass and cross-mass matrices

The boundary mass matrix $\boldsymbol{W}$ of a side set is the consistent facet mass matrix (5) on the reference coordinates, with the default facet quadrature of each facet type (three points for a linear triangle, four for a bilinear quadrilateral, seven for a quadratic triangle). The cross-mass matrix of a destination side $d$ and a source side $s$ of a shared interface,

$$L^{ds}_{mn} = \sum_{\text{facets of } d} \sum_{q} \varphi^d_m(\boldsymbol{x}_q)\, \varphi^s_n(\boldsymbol{\xi}_s(\boldsymbol{x}_q))\, J_q\, w_q, \tag{16}$$

is integrated over the facets of the destination, each quadrature point located on the source by the projection of Sec. 5.1; the matrix integrated over the source facets is $(L^{sd})^T$. Two projections are built from these matrices:

$$\text{Dirichlet projector} \quad \boldsymbol{P}_d = \boldsymbol{W}_d^{-1} \boldsymbol{L}, \tag{17}$$

$$\text{Neumann projector} \quad \boldsymbol{N}_d = \boldsymbol{L}\, \boldsymbol{W}_s^{-1}, \tag{18}$$

$\boldsymbol{P}_d$ mapping nodal values of the source to the $L^2$ projection of their trace onto the destination, and $\boldsymbol{N}_d$ mapping nodal forces of the source to nodal forces of the destination. For $\boldsymbol{P}_d$ the code prefers the source-integrated $\boldsymbol{L}$ and uses the destination-integrated one instead when $\max|\boldsymbol{P}_d\mathbf{1}-\mathbf{1}| > 10^{-10}$; for $\boldsymbol{N}_d$ it prefers the destination-integrated $\boldsymbol{L}$ and uses the other instead when the relative error of the column sums of $\boldsymbol{L}$ against $\boldsymbol{W}_s\mathbf{1}$ exceeds $10^{-10}$. The classical Dirichlet–Neumann, Robin–Robin, and contact couplings build $\boldsymbol{P}_d$ and $\boldsymbol{N}_d$ independently on each side by these rules. The matching exchange instead builds both sides' transfers from one cross-mass matrix $\boldsymbol{B}$, integrated over the facets of the side with more interface nodes, with no alternative: $\Pi_1 = \boldsymbol{W}_1^{-1}\boldsymbol{B}$ and $\Pi_2 = \boldsymbol{W}_2^{-1}\boldsymbol{B}^T$, each side's force transfer the transpose of the partner's kinematic transfer, and the partition-of-unity error $\max(\max|\Pi_1\mathbf{1}-\mathbf{1}|, \max|\Pi_2\mathbf{1}-\mathbf{1}|)$ logged. Only $\Pi_D$ and its transpose enter the matching exchange; the projection onto the other side is never applied, which matters on curved or non-nested interfaces where it is inaccurate (0.72 on the cylinders of Sec. 11, $1.2\times10^{-2}$ on the beam at mesh ratio 1:0.75, against $10^{-15}$ for $\Pi_D$).

**Displacement, not position.** A Dirichlet–Neumann transfer projects the partner's displacement, $\boldsymbol{P}_d\boldsymbol{u}_s$. Projecting the current position $\boldsymbol{P}_d(\boldsymbol{X}_s+\boldsymbol{u}_s)$ and subtracting the destination's reference position $\boldsymbol{X}_d$ is not equivalent on a curved interface, because $\boldsymbol{P}_d\boldsymbol{X}_s \neq \boldsymbol{X}_d$ by the sagitta of the facets between the two meshes; that difference is a spurious displacement of the size of the chord error. The overlapping and contact couplings transfer the position, as their formulation requires (Secs. 6 and 10).

# 6 The overlapping Dirichlet coupling

Two subdomains overlap; each has a side set $\Gamma_k$ of its own boundary that lies inside the partner. The coupling is a Dirichlet condition on $\Gamma_k$ with the datum interpolated from the partner's solution; both sides are Dirichlet, there is no force transfer, and no role is assigned.

**Strong form (the default).** For each node $i$ of $\Gamma_k$, the element of the partner's `source block` that contains its reference position $\boldsymbol{X}_i$ and the shape function values $N_a$ at that point are found once (Sec. 5.1). At every application the condition prescribes, with the partner fields interpolated in time by (10),

$$\boldsymbol{u}_i = \sum_a N_a\,(\boldsymbol{X}_a^j + \boldsymbol{u}_a^j) - \boldsymbol{X}_i, \qquad \boldsymbol{v}_i = \sum_a N_a\,\boldsymbol{v}_a^j, \qquad \boldsymbol{a}_i = \sum_a N_a\,\boldsymbol{a}_a^j, \tag{19}$$

and marks the three components of node $i$ not free. The displacement is the interpolated current position minus the node's own reference position, which equals the interpolated displacement up to the residual of the parametric inversion.

**Weak form (`weak: true`).** The datum is the $L^2$ projection of the partner's field onto the trace space of $\Gamma_k$: $\boldsymbol{P} = \boldsymbol{W}_k^{-1}\boldsymbol{L}$, with $\boldsymbol{W}_k$ the boundary mass matrix of $\Gamma_k$ and $\boldsymbol{L}$ the rectangular matrix of (16) whose quadrature points on the facets of $\Gamma_k$ are located in the partner's volume elements, so that $\boldsymbol{L}$ has one column per node of the partner. The condition prescribes $\boldsymbol{u}|_{\Gamma_k} = \boldsymbol{P}(\boldsymbol{X}^j+\boldsymbol{u}^j) - \boldsymbol{X}|_{\Gamma_k}$, $\boldsymbol{v}|_{\Gamma_k} = \boldsymbol{P}\boldsymbol{v}^j$, $\boldsymbol{a}|_{\Gamma_k} = \boldsymbol{P}\boldsymbol{a}^j$.

**Properties.** The three fields are prescribed in both the dynamic and the quasi-static case; a quasi-static subdomain uses only the displacement, and a Newmark subdomain with a consistent mass receives the prescribed acceleration in its free rows through the coupling term of the mass

matrix. The element search and the shape function values are fixed at construction on the reference configuration and are not updated for finite deformation. The datum is not relaxed: the relaxation keys have no effect on this coupling, and the interface predictor does not apply to it. The iteration stops on the displacement update of Sec. 3.3.

### 6.1 Overlap $L^2$ error

With `compute overlap L2 relative error: disp|velo|acce` the coupling records, for the set $\mathcal{O}$ of nodes of subdomain $k$ that lie in the partner's `source block` (the nodes of elements whose centroid, or any node, is found there, restricted to those found themselves), the relative nodal error of the selected field $\boldsymbol{\phi}$,

$$e_k = \Big(\sum_{i\in\mathcal{O}} \big\|\boldsymbol{\phi}_i - \sum_a N_a(\boldsymbol{\xi}_i)\,\boldsymbol{\phi}_a^j\big\|^2 \Big/ \sum_{i\in\mathcal{O}} \|\boldsymbol{\phi}_i\|^2\Big)^{1/2}, \tag{20}$$

an unweighted nodal norm of the current fields, reported at the end of every stop that took at least one Schwarz iteration and written to `overlap-l2-<field>-rel-errors-<stop>.csv` at the CSV output times. It is also the criterion of the overlap $L^2$ swap of Sec. 3.8.

### 6.2 Blended energy over the overlap

The overlap region is meshed twice, so the sum of the subdomain energies counts it twice. The blended energy output weights the energy density at each quadrature point $\boldsymbol{x}$ of subdomain $k$ by the Arlequin partition of unity

$$w_k(\boldsymbol{x}) = \frac{d_k(\boldsymbol{x})}{d_k(\boldsymbol{x}) + d_j(\boldsymbol{x})} \tag{21}$$

when $\boldsymbol{x}$ lies in the partner $j$, and $w_k = 1$ otherwise, with $d_k$ the distance from $\boldsymbol{x}$ to the overlap side set $\Gamma_k$ of subdomain $k$ and $d_j$ the distance to $\Gamma_j$, both on the reference configuration and each the minimum over the facets of the side set of the distance to the closest point of the facet. Membership is decided by locating $\boldsymbol{x}$ in the partner mesh (all blocks) with the `search tolerance`, so the meshes need not conform. For two subdomains with mutual overlap conditions $w_k + w_j = 1$ in the overlap; with several partners $d_j$ is the sum over the partners containing $\boldsymbol{x}$ and $d_k$ the minimum over the own overlap side sets. An overlap in which $d_k + d_j$ is below $10^{-3}$ of the element size at some point (a zero-width overlap) aborts the run. The weights are evaluated once and cached. Nonoverlapping and contact decompositions take unit weights, so their total is the plain sum. The weighted stored and kinetic energies, the latter in the staggered form for central difference, are written at every stop as described in Sec. 3.7, and reduce to the energy of the undecomposed body at Schwarz convergence.

## 7 The classical Dirichlet–Neumann exchange

Two subdomains share an interface $\Gamma$ given on each side as a side set. The side whose `default BC type` is `Dirichlet` receives the partner's kinematic fields and the other, `Neumann`, receives the partner's internal force; the two values must differ. With `swap BC types: true` on both sides the roles are exchanged once per stop, before its Schwarz iteration.

**Dirichlet side.** With $\boldsymbol{P}_D$ the Dirichlet projector (17) of side $D$ and the partner fields interpolated in time, relaxed as in Sec. 4,

$$\boldsymbol{u}_D|_\Gamma = \boldsymbol{P}_D\,\boldsymbol{u}_N, \qquad \boldsymbol{v}_D|_\Gamma = \boldsymbol{P}_D\,\boldsymbol{v}_N, \qquad \boldsymbol{a}_D|_\Gamma = \boldsymbol{P}_D\,\boldsymbol{a}_N, \tag{22}$$

all three prescribed and their degrees of freedom marked not free.

**Neumann side.** With $\boldsymbol{N}_N$ the Neumann projector (18) of side $N$, the force

$$\boldsymbol{f}_N = \boldsymbol{N}_N\big(-\,\boldsymbol{f}_D^{\mathrm{int}}|_\Gamma\big) \tag{23}$$

is added to the boundary force of the interface nodes of $N$, with $\boldsymbol{f}_D^{\mathrm{int}}$ interpolated in time from the partner's history. This is the static reaction of the Dirichlet side; it omits the inertia of the interface nodes and the loads applied on them, which Sec. 11.8 measures to be of the size of the reaction itself in dynamics.

**Relaxation.** Under fixed and recursive Aitken relaxation the three partner fields are blended with the same factor, formed from the partner's whole displacement field. Under secant Aitken relaxation the displacement is blended and the velocity and acceleration are recovered from the partner's Newmark predictors by (4) when the partner is a Newmark subdomain, and left unrelaxed otherwise. Under Anderson acceleration the three fields have separate histories and share the coefficients formed from the projected displacement residual.

**Diagnostics and stopping.** At every Schwarz iteration the log reports, for each Dirichlet–Neumann pair, the one-sided jumps of the velocity and the displacement and the force residual defined in Sec. 8.5; for the classical exchange they are reported only. The iteration stops on the displacement update of Sec. 3.3, and the initial acceleration of each subdomain is computed per subdomain with the interface rows excluded from the correction (Sec. 3.6).

### 7.1 Interface predictor

With `interface predictor: true` the first-listed subdomain of a nonoverlapping pair (Dirichlet–Neumann or Robin–Robin) reads at iteration 0 a predictor of the partner's state instead of the state at $t_{s-1}$. For a quasi-static pair each side extrapolates its interface displacement, $\boldsymbol{u}^{\mathrm{pred}} = 2\boldsymbol{u}_s - \boldsymbol{u}_{s-1}$, from the second stop on. For a dynamic pair the predictor velocity is the velocity of a perfectly inelastic collision of the two interface traces, with the row sums of the boundary mass matrices as masses, $\boldsymbol{v}_k^{\mathrm{pred}} = (\boldsymbol{m}_k\boldsymbol{v}_k + \boldsymbol{P}_k\boldsymbol{m}_j \circ \boldsymbol{P}_k\boldsymbol{v}_j)/(\boldsymbol{m}_k + \boldsymbol{P}_k\boldsymbol{m}_j)$ on side $k$ and its projection on side $j$, the predictor displacement is $\boldsymbol{u}_s + \Delta t_c\,\boldsymbol{v}^{\mathrm{pred}}$, and the acceleration is that of the stop. The predicted internal force of the side whose force is read is $2\boldsymbol{f}_s - \boldsymbol{f}_{s-1}$.

**Restrictions.** The exchange does not conserve the energy of a dynamic problem (Sec. 11.8); the matching exchange is the dynamic coupling.

## 8 The matching Dirichlet–Neumann exchange

The matching exchange is selected by `matching: velocity` or `matching: displacement` on both sides of a `Schwarz DN nonoverlap` pair, with the same value on both. It is called the matching exchange because its Dirichlet side matches one kinematic quantity of the partner, the velocity or,

for trapezoidal pairs, the displacement, and derives the other two from its own time integrator; the coupled problem it solves is the constrained problem of Sec. 8.1, whose constraint is that matching and whose multiplier is the interface force. It differs from the classical exchange in five respects: the Neumann side receives the full dynamic (d'Alembert) reaction of its partner; both transfers are built from one cross-mass matrix, so that the force transfer is the transpose of the kinematic transfer; the Dirichlet side receives one quantity only; the iteration stops on the interface jump and the force residual; and the initial acceleration is computed on the coupled system.

## 8.1 The constrained coupled problem

The coupled problem consists of (1) on both sides with one interface force acting in opposite directions,

$$\boldsymbol{\lambda}_D = \boldsymbol{\lambda}, \qquad \boldsymbol{\lambda}_N = -\Pi_D^T \boldsymbol{\lambda}, \tag{24}$$

and the velocity constraint at the end of every window,

$$\boldsymbol{v}_D = \Pi_D \, \boldsymbol{v}_N \quad \text{on } \Gamma. \tag{25}$$

The interface force $\boldsymbol{\lambda}$ is the Lagrange multiplier of (25). On conforming meshes $\Pi_D$ is the identity on the interface nodes and the formulation is that of Gravouil and Combescure [2], a dual Schur formulation with velocity continuity. Prakash and Hjelmstad [6] impose the constraint only at the end of a coarse step, with the force on the fine side interpolated linearly in time; Connors, Owen, Kuberry, and Bochev [3] cast the same structure as a discontinuous-Galerkin-in-time multirate framework whose lowest flux order is this scheme. All three assume the same spatial discretization on both sides of the interface, so that their interface map is Boolean; the matching exchange removes that assumption, with the $L^2$ projector $\Pi_D$ in place of the Boolean map, and the identities of Sec. 8.2 hold for any cross-mass matrix $\boldsymbol{B}$.

The force on the interface rows of side $D$ is, by (1), its d'Alembert reaction

$$\boldsymbol{r}_D = \boldsymbol{C}_D\big(\boldsymbol{M}_D \, \boldsymbol{a}_D + \boldsymbol{f}_D^{\text{int}} - \boldsymbol{f}_D^{\text{body}} - \boldsymbol{f}_D^{\text{bdry}}\big) = \boldsymbol{\lambda}, \tag{26}$$

with $\boldsymbol{f}_D^{\text{body}}$ the body force and $\boldsymbol{f}_D^{\text{bdry}}$ the other boundary loads acting on the interface rows. The reaction includes the inertia of the interface nodes; the static reaction $-\boldsymbol{f}^{\text{int}}$ omits a term of the size of the reaction itself (Sec. 11.8). The mass in (26) is the consistent mass for a Newmark side and the lumped mass for a central difference side.

## 8.2 The energy identity

Connors et al. (their Eq. (102)) give the energy balance of one side. For side $k$ advanced by $m_k$ Newmark steps with a linear stiffness, an interface force $\boldsymbol{f}_k$ on its interface rows, and no other loads,

$$\tilde{E}_k^{m_k} - \tilde{E}_k^0 + \Big(\gamma_k - \tfrac{1}{2}\Big) \sum_{n=1}^{m_k} (\Delta \boldsymbol{a}_k^n)^T \boldsymbol{S}_k \, \Delta \boldsymbol{a}_k^n = \frac{1}{\Delta t_k} \sum_{n=1}^{m_k} (\Delta \boldsymbol{v}_k^n)^T \boldsymbol{C}_k^T \, \Delta \boldsymbol{f}_k^n, \tag{27}$$

with $\Delta(\cdot)^n = (\cdot)^n - (\cdot)^{n-1}$ and $\tilde{E}_k$ the term of side $k$ in (9). With $\gamma_k = 1/2$ the second term vanishes.

**Proposition 1** (Equal steps)**.** *Let both sides take one step $\Delta t$ per window, with $\gamma = 1/2$ on both, a linear stiffness, no loads on the interface rows other than the interface force, $\boldsymbol{f}_D = \boldsymbol{\lambda}$ and $\boldsymbol{f}_N = -\Pi_D^T \boldsymbol{\lambda}$, and $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ at every step. Then $\tilde{E}_D + \tilde{E}_N$ is constant.*

*Proof.* By (27) the change over one step is $\Delta t^{-1}\big[(\Delta \boldsymbol{v}_D)^T \Delta\boldsymbol{\lambda} - (\Delta \boldsymbol{v}_N)^T \Pi_D^T \Delta\boldsymbol{\lambda}\big] = \Delta t^{-1}(\Delta \boldsymbol{v}_D - \Pi_D \Delta \boldsymbol{v}_N)^T \Delta\boldsymbol{\lambda} = 0$. □

The cancellation needs only the transposed force transfer and the one-sided constraint; it holds for any cross-mass matrix $\boldsymbol{B}$, so for any mesh ratio, and for any pair of Newmark schemes with $\gamma = 1/2$. It is the interface work identity of Gravouil and Combescure (their Eq. (49)) written with $\Pi_D$ in place of the Boolean interface map. Identical integrators at equal steps conserve $E$ as well: the constraint and the common Newmark relations then give $\boldsymbol{u}_D = \Pi_D \boldsymbol{u}_N$ up to the initial data, and the interface power $\boldsymbol{\lambda}^T(\boldsymbol{v}_D - \Pi_D \boldsymbol{v}_N)$ vanishes at the end of every step.

**Proposition 2** (Different steps)**.** *Let side $D$ take one step $\Delta T$ per window and side $N$ take $m$ steps $\Delta T/m$, under the conditions of Proposition 1 except that $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ holds only at the two ends of the window, and let the force on $N$ at its substep $j$ be the linear interpolation*

$$\boldsymbol{f}_N^{\,j} = -\Pi_D^T\Big[\big(1 - \tfrac{j}{m}\big)\boldsymbol{\lambda}^0 + \tfrac{j}{m}\,\boldsymbol{\lambda}^1\Big], \qquad j = 0, \ldots, m, \tag{28}$$

*of the forces $\boldsymbol{\lambda}^0$ and $\boldsymbol{\lambda}^1$ on $D$ at the window ends. Then $\tilde{E}_D + \tilde{E}_N$ is constant over the window.*

*Proof.* $\Delta \boldsymbol{f}_N^{\,j} = -\Pi_D^T \Delta\boldsymbol{\lambda}/m$ for every substep, with $\Delta\boldsymbol{\lambda} = \boldsymbol{\lambda}^1 - \boldsymbol{\lambda}^0$, so the right side of (27) for $N$ is $\frac{m}{\Delta T}\sum_{j=1}^{m} (\Delta \boldsymbol{v}_N^{\,j})^T(-\Pi_D^T \Delta\boldsymbol{\lambda}/m) = -\Delta T^{-1}\big(\Pi_D(\boldsymbol{v}_N^m - \boldsymbol{v}_N^0)\big)^T \Delta\boldsymbol{\lambda}$, and that of $D$ is $\Delta T^{-1}(\boldsymbol{v}_D^1 - \boldsymbol{v}_D^0)^T \Delta\boldsymbol{\lambda}$. The two cancel when $\boldsymbol{v}_D = \Pi_D \boldsymbol{v}_N$ at both window ends. □

This is Eq. (103) of Connors et al. with $\Pi_D$ in place of their Boolean map, their condition (108) at flux order $r = 1$, and Eq. (70) of Prakash and Hjelmstad. The force on $N$ at the window start must be the converged force of the previous window, and the loads on the interface rows of $D$ must be constant within the window, because the reaction (26) is formed from their current value.

**The coarse step must be the Dirichlet side.** The same sums cancel when the fine-step side is the Dirichlet side and meets the linearly interpolated coarse velocity at every substep, the arrangement of Gravouil and Combescure. That arrangement conserves $\tilde{E}$ but lacks the flux projection property of Connors et al. (their Eq. (80)): the impulse that the fine side receives over the window, the trapezoidal sum of its reactions at every substep, differs from the impulse that the coarse side receives, the trapezoid of the two end values. The interface impulse residual of a window,

$$\boldsymbol{\rho} = \sum_{j=1}^{m} \frac{\Delta T}{2m}\big(\boldsymbol{f}_N^{\,j-1} + \boldsymbol{f}_N^{\,j}\big) + \Pi_D^T \frac{\Delta T}{2}\big(\boldsymbol{\lambda}^0 + \boldsymbol{\lambda}^1\big), \tag{29}$$

vanishes identically in the coarse-step arrangement. In the fine-step arrangement it is of the order of the impulse itself, and the energy $E$ varies by 12 to 51% over the period of the first bending mode of the beam while $\tilde{E}$ is conserved (Sec. 11.3). $\tilde{E}$ weights the lowest mode least, so it does not detect the variation. The code does not enforce the assignment; it is a rule for the input (Sec. 8.7).

**Scope.** Propositions 1 and 2 hold for a linear stiffness. For a nonlinear material the exchange still satisfies the velocity constraint exactly and does no interface work at the fixed point; the energy then varies by the time-step error of the subdomain integrators themselves. For central difference that error is of order $\Delta t^2$ and does not grow: the scheme conserves a modified energy that differs from $E$ by a state-dependent term of order $\Delta t^2$ [8]. Conservation of $E$ to roundoff for a nonlinear material requires an energy-momentum integrator [9, 10] in each subdomain and a midpoint velocity constraint in the exchange, which the code does not have.

### 8.3 Setup

1. Assign the Dirichlet role (Sec. 8.7) with `default BC type`.
2. Assemble $\boldsymbol{B}$, $\boldsymbol{W}_D$, and $\boldsymbol{W}_N$ of (5) on the reference configuration with the facet quadrature of the side with more interface nodes, and form $\Pi_D$ of (6) (Sec. 5.2). Check that $\Pi_D$ reproduces constants (the partition-of-unity error $\max|\Pi_D\mathbf{1}-\mathbf{1}|$ is at roundoff when the quadrature is exact).
3. Compute the initial acceleration on the coupled system (Sec. 8.6).

Both integrators must be Newmark or central difference without HHT-$\alpha$. Displacement matching requires the same Newmark parameters with $\beta > 0$ and the same step on both sides, and aborts if the steps taken differ during the run; velocity matching admits different steps and central difference. The exchange cannot be combined with `swap BC types`.

### 8.4 One window

Given the converged state at the window start and the force $\boldsymbol{\lambda}^0$ of the previous window:

1. Initialize the datum $\boldsymbol{q}_D^*$ of the Dirichlet side, where $\boldsymbol{q}$ is the matched quantity, with the projected partner value at the window start (the first iteration of a stop reads the partner's state at $t_{s-1}$, Sec. 3.2).
2. Repeat until the stopping criterion of Sec. 8.5 holds:
   (a) Advance side $D$ over the window with $\boldsymbol{q}_D = \boldsymbol{q}_D^*$ imposed on its interface nodes at the end of each of its substeps, the other two kinematic fields from (3) (velocity matching) or (4) (displacement matching) with its own $\beta$, $\gamma$, and substep, and form its reaction $\boldsymbol{\lambda}^1 = \boldsymbol{r}_D$ by (26).
   (b) Advance side $N$ over its substeps with the force (28) on its interface nodes (at equal steps, $\boldsymbol{f}_N = -\Pi_D^T\boldsymbol{\lambda}^1$); in the code this is the time interpolation (10) of $D$'s acceleration and internal force history in (26), with $D$'s current body and boundary forces.
   (c) Form the new datum $\Pi_D\boldsymbol{q}_N$ at the window end, with $\boldsymbol{q}_N$ interpolated in time to each substep of $D$ under subcycling, and relax it (Sec. 4, with the Aitken factor and the Anderson coefficients formed from the projected trace) into $\boldsymbol{q}_D^*$.
3. Commit the window and keep $\boldsymbol{\lambda}^1$ as the next $\boldsymbol{\lambda}^0$.

Each Schwarz iteration restores both sides to the state at the window start before it advances them. The datum is imposed before the Newmark predictor of the receiving substep, with the interface rows still holding the substep-start state from which the predictors of (3) are formed; at $t = 0$, where no step is in progress, the Dirichlet side imposes the three projected fields instead. The velocity constraint is the general choice; the displacement constraint, between the same trapezoidal scheme on both sides at equal steps, gives the same solution to the Schwarz tolerance (test 136).

### 8.5 Stopping criterion

With $\|\boldsymbol{x}\|_{\boldsymbol{W}}^2 = \sum_c \boldsymbol{x}_c^T\boldsymbol{W}\boldsymbol{x}_c$ over the three Cartesian components, the Dirichlet side's one-sided relative jump of a quantity $\boldsymbol{q}$ and the force residual are

$$\frac{\|\boldsymbol{q}_D - \Pi_D\boldsymbol{q}_N\|_{\boldsymbol{W}_D}}{\|\boldsymbol{q}_D\|_{\boldsymbol{W}_D}}, \qquad \frac{\|\boldsymbol{f}_N + \Pi_D^T\boldsymbol{r}_D\|_{\boldsymbol{W}_N^{-1}}}{\|\Pi_D^T\boldsymbol{r}_D\|_{\boldsymbol{W}_N^{-1}}}, \tag{30}$$

the discrete $L^2$ norms of the trace and of its dual, with $\boldsymbol{f}_N$ the force applied on $N$ in the same iteration. The log reports the velocity and displacement jumps and the force residual of every Dirichlet–Neumann pair at every Schwarz iteration; for a matching pair each value is the largest over the substeps of the window, the jumps at every substep of $D$ against $N$'s history interpolated to that time, the force residual at every substep of $N$ against $D$'s history. The iteration of a matching pair stops when the jump of the matched quantity and the force residual are both at or below $\epsilon$, the `matching relative tolerance`, or when the root mean square jump $\|\boldsymbol{q}_D - \Pi_D \boldsymbol{q}_N\|_{\boldsymbol{W}_D}/(\mathbf{1}^T \boldsymbol{W}_D \mathbf{1})^{1/2}$ is at or below the `matching absolute tolerance` (default 0, no absolute test), which is in the units of the matched quantity; a displacement tolerance divided by the step is not a valid velocity floor (on the cylinders of Sec. 11 it amounts to 200 m/s and stops the iteration after one pass). When every coupling of the simulation is a matching pair, this is the only stopping test, also at the first iteration of a stop, and the displacement update of Sec. 3.3 is not used. Measured against the earlier rule that also required the displacement update to meet the controller's absolute tolerance at the first iteration, this changes only the stops at which nothing has yet reached the interface: on the nonconforming beam (2000 stops) the 7953 Schwarz iterations and the energy output are identical, and on the linear cylinders (2000 stops, fixed relaxation 0.5, $\epsilon = 10^{-12}$) the first seven stops take 0 iterations instead of 1, every later count of the 53 694 is the same, and the energies of the two runs differ by at most $7.7 \times 10^{-14}$ ($E$) and $4.4 \times 10^{-12}$ ($\tilde{E}$).

**Stall.** Let $\mu_i$ be the larger of the jump and the force residual, maximized over the matching pairs, at iteration $i$. The iteration is stalled when $\mu_i$ has not fallen 5% below its smallest value for three consecutive counted iterations (every iteration when all couplings are matching pairs; otherwise only those at which the displacement update has converged). The residual is not monotone under Aitken or Anderson acceleration, so a single non-decreasing iteration is not a stall. A stalled stop is accepted with a warning and the Anderson history cleared (`stalled interface jump action: warn`, the default) or aborts the run (`abort`). Every stopping tolerance leaves a drift of the conserved quantity of the order of $\epsilon$ per window, accumulated over the windows of the run (Sec. 11.4).

### 8.6 Coupled initial acceleration

The initial accelerations must satisfy the coupled problem at $t = 0$, including the acceleration constraint $\boldsymbol{a}_D = \Pi_D \boldsymbol{a}_N$, or the first step changes $\tilde{E}$ by 0.2 to 0.9% and, for an explicit pair, the difference of the two interface accelerations alternates in sign at every step without decaying. The coupled initial acceleration is a Schwarz iteration at $t = 0$ in which $D$ imposes the projected partner displacement, velocity, and acceleration, $N$ receives the d'Alembert reaction of $D$, and each side solves for its acceleration with the coupling rows included in the correction of Sec. 2. The imposed interface acceleration is relaxed with the fixed factor, with the secant Aitken factor (14) under either Aitken option, or by Anderson acceleration, as the input selects. The iteration stops when the one-sided acceleration jump and the force residual of every pair are at or below $\epsilon$, or after `maximum iterations` with a warning. It converges in 2 iterations on conforming meshes with identical integrators, 16 to 24 on nonconforming meshes, and about 30 for mixed pairs at a fixed relaxation factor of 0.5, and in 2 to 9 with Anderson acceleration. For two explicit sides the direct solve of Sec. 8.8 gives it in one step.

### 8.7 Assignment of the Dirichlet role

The linearized gain of one Schwarz iteration at fixed relaxation is

$$\boldsymbol{G} = -\,\Pi_D\, \boldsymbol{H}_N\, \Pi_D^T\, \boldsymbol{S}_D, \tag{31}$$

with $\boldsymbol{H}_N = \boldsymbol{C}_N \tilde{\boldsymbol{M}}_N^{-1} \boldsymbol{C}_N^T$ the interface flexibility of $N$ ($\tilde{\boldsymbol{M}} = \boldsymbol{M} + \beta \Delta t^2 \boldsymbol{K}$, the lumped mass for central difference) and $\boldsymbol{S}_D$ the interface dynamic stiffness of $D$. A fixed relaxation factor $\theta$ converges if and only if the spectral radius of $(1-\theta)\boldsymbol{I} + \theta \boldsymbol{G}$ is below one; for a real spectrum in $[a, b]$ the optimal factor is $\theta^* = 2/((1-a)+(1-b))$ with contraction $(b-a)/((1-a)+(1-b))$. Table 2 lists the computed eigenvalues on the beam. The rules that follow, in order of precedence:

1. With different time steps, the coarse-step side is the Dirichlet side (Proposition 2 and the impulse residual).
2. With different integrators, the implicit side is the Dirichlet side: with the explicit side as the Dirichlet side the eigenvalues reach $-13.3$ and every factor $\theta \geq 0.5$ diverges.
3. With the same integrator on a nonconforming interface, the finer mesh is the Dirichlet side: the eigenvalues then lie in $[g, 0]$ with $|g| \leq 1$; with the coarser mesh as the Dirichlet side they reach $-8$ for an explicit pair.

When rules 1 and 3 conflict (a finer mesh with a finer step), giving the finer mesh the coarser step satisfies both where the stability limit of an explicit side allows it. With Anderson acceleration every assignment measured converges, so the rules then set the iteration count, not the feasibility.

### 8.8 Direct interface solve for two explicit subdomains

For central difference on both sides at equal steps, with velocity matching, the iteration can be replaced by one solve per step (`interface solve: direct` on both sides; every coupling of the simulation must then be such a pair). Central difference computes $\boldsymbol{u}^{n+1}$ before the acceleration, and the internal force at $\boldsymbol{u}^{n+1}$ does not depend on the interface force. Advancing both sides with free interface rows and no interface force gives the velocities $\boldsymbol{v}^{\text{free}}$; the interface force adds $\gamma \Delta t\, \boldsymbol{m}_D^{-1} \boldsymbol{\lambda}$ to the velocity of $D$ and $-\gamma \Delta t\, \boldsymbol{m}_N^{-1} \Pi_D^T \boldsymbol{\lambda}$ to that of $N$, with $\boldsymbol{m}_D$, $\boldsymbol{m}_N$ the lumped masses of the interface rows. The velocity constraint then gives, per Cartesian component,

$$\gamma \Delta t \big( \boldsymbol{m}_D^{-1} + \Pi_D\, \boldsymbol{m}_N^{-1}\, \Pi_D^T \big) \boldsymbol{\lambda} = \Pi_D \boldsymbol{v}_N^{\text{free}} - \boldsymbol{v}_D^{\text{free}}. \tag{32}$$

The matrix is symmetric positive definite, of the size of the Dirichlet trace, and constant while the lumped masses and $\Pi_D$ are; rows held by other Dirichlet conditions take a zero inverse mass, and a row held on both sides makes the matrix singular and aborts. It is the interface operator $\boldsymbol{H} = \sum_k \gamma_k \Delta t_k \boldsymbol{C}_k \tilde{\boldsymbol{M}}_k^{-1} \boldsymbol{C}_k^T$ of Gravouil and Combescure for two lumped-mass sides, written on the Dirichlet trace. It is factored once by a dense Cholesky factorization (3920 unknowns per component on the cylinders, 0.7 s), and at every step both sides are advanced, (32) is solved, and the interface accelerations and velocities are corrected by $\boldsymbol{m}_D^{-1} \boldsymbol{\lambda}$ and $\gamma \Delta t\, \boldsymbol{m}_D^{-1} \boldsymbol{\lambda}$ on $D$, and by the same with $-\Pi_D^T \boldsymbol{\lambda}$ and $\boldsymbol{m}_N$ on $N$; the displacements are not corrected, because central difference has already formed them. The solve is exact for nonlinear materials, because the internal force is evaluated at the predicted displacement. The same matrix gives the coupled initial acceleration from $\boldsymbol{a}_D = \Pi_D \boldsymbol{a}_N$. Relaxation and the stopping tolerances do not apply; the stop takes no Schwarz iteration.

### 8.9 Requirements on a subdomain code

The exchange requires of each subdomain code, beyond a standard step:

1. a step with a prescribed velocity (or displacement) on a set of boundary nodes, with the other two kinematic fields of those nodes from (3) or (4) (the Dirichlet side);

2. a step with a prescribed nodal force on a set of boundary nodes, constant or linear in time over the window (the Neumann side);
3. the nodal d'Alembert reaction (26) on a set of boundary nodes after a step;
4. saving and restoring the state at the window start;
5. the boundary mass matrix and the facet quadrature of its interface, to assemble $\boldsymbol{B}$ (performed once, by either code or by the coupler).

An implicit code provides items 1 to 3 as a Newton solve at fixed boundary data; an explicit code provides them as a division by the lumped mass. The coupler itself holds $\Pi_D$, the relaxation state, and, for two explicit sides, the factorization of (32). The other couplings of this note require subsets of the same operations: the overlapping coupling items 1 (with all three fields prescribed) and 4 and a point location in the partner mesh; the classical exchange items 1, 2, 4, and 5 and the internal force on the interface nodes; the Robin–Robin condition a step with the term $\alpha \boldsymbol{W}\boldsymbol{u}$ and its tangent added to the residual.

# 9 The Robin–Robin condition

Two subdomains share an interface. Each side $k$ imposes the Robin condition $\boldsymbol{t}_k + \alpha_k \boldsymbol{W}_k \boldsymbol{u}_k = \boldsymbol{g}_k$ on its interface rows, with $\boldsymbol{t}_k$ its interface traction, $\boldsymbol{W}_k$ its boundary mass matrix, and $\alpha_k > 0$ its `robin parameter` (required; the two sides may differ). The datum is formed from the partner's traction and displacement,

$$\boldsymbol{g}_k = \boldsymbol{N}_k\big(-\boldsymbol{f}_j^{\mathrm{int}}|_\Gamma\big) + \alpha_k \boldsymbol{W}_k \boldsymbol{P}_k \boldsymbol{u}_j, \tag{33}$$

with $\boldsymbol{P}_k$ and $\boldsymbol{N}_k$ the Dirichlet and Neumann projectors (17) and (18) of side $k$, each side building its own, and the partner fields interpolated in time. The partner's internal force is its elastic internal force; the partner's own Robin term is not part of it. The datum is added to the boundary force of the interface nodes of $k$; the self term $\alpha_k \boldsymbol{W}_k \boldsymbol{u}_k|_\Gamma$ is added to the internal force at every residual evaluation, including the initial-acceleration solves, and its tangent $\alpha_k \boldsymbol{W}_k$ is added to the Hessian of implicit solvers. The condition solved on side $k$ is therefore

$$\boldsymbol{f}_k^{\mathrm{int}}(\boldsymbol{u}_k) + \alpha_k \boldsymbol{W}_k \boldsymbol{u}_k|_\Gamma - \boldsymbol{f}_k^{\mathrm{ext}} = \boldsymbol{N}_k\big(-\boldsymbol{f}_j^{\mathrm{int}}|_\Gamma\big) + \alpha_k \boldsymbol{W}_k \boldsymbol{P}_k \boldsymbol{u}_j. \tag{34}$$

**Relaxation.** The side listed later in `domains` relaxes its datum: $\boldsymbol{\lambda}_i = (1-\theta_i)\,\boldsymbol{\lambda}_{i-1} + \theta_i\,\boldsymbol{g}_i$, with the fixed, Aitken, or Anderson factor of Sec. 4 formed from the datum itself, and $\boldsymbol{\lambda}_{-1} = 0$ at a new slot, so that the first iteration of every stop applies $\theta_0 \boldsymbol{g}_0$ on that side. The earlier-listed side applies $\boldsymbol{g}_i$ unrelaxed.

**Initialization and stopping.** When an interface node has a nonzero initial displacement or velocity, the initialization runs the two refinement passes of Sec. 3.6. The iteration stops on the displacement update of Sec. 3.3. The condition is not absorbing: under a dynamic integrator it can increase the energy, and the code warns when a Robin–Robin side is not quasi-static. It is the coupling of the quasi-static examples; no impedance (velocity-proportional) term is implemented.

# 10 The contact coupling

Two bodies whose surfaces, given as a side set on each, may come into contact are coupled by a Dirichlet–Neumann exchange that is active while contact is detected. Each side names its own side

set, the partner, and the partner's side set, and a `friction type`: `frictionless` or `tied`. The side of the subdomain listed first in `domains` is the Dirichlet side; `swap BC types: true` exchanges the roles at every stop in which contact is active. The transfer operators are the Dirichlet and Neumann projectors of Sec. 5.2, built once at initialization on the reference configuration.

**Detection.** One flag holds the contact state of the whole simulation. It is evaluated after every stop has been solved, and at initialization. While contact is inactive, the stop is solved with the subdomains advanced independently and the contact surfaces traction-free, and contact becomes active when any node $\boldsymbol{x}$ of the side set of a contact condition, in the current configuration, has a closest-point projection $\boldsymbol{y}$ onto a partner facet, within a search radius of ten times the smallest nodal spacing of its own side set, that lies inside the facet and on the far side of it: the signed distance $-\operatorname{sign}((\boldsymbol{y}-\boldsymbol{x})\cdot\boldsymbol{n})\,\|\boldsymbol{y}-\boldsymbol{x}\| \le 0$, with $\boldsymbol{n}$ the facet normal. While contact is active, the stop is solved by the Schwarz iteration, and contact persists while any node of a contact side set has a compressive normal force, $\boldsymbol{f}_i \cdot \boldsymbol{n}_i \le 0$, with $\boldsymbol{f}_i$ the force transferred from the partner and $\boldsymbol{n}_i$ the nodal normal of the side set. When the state changes after a stop, the stop is restored and solved once more in the new state, without a further check. The state is written to `contact-<stop>.csv` at the CSV output stops.

**Dirichlet side.** With $\boldsymbol{P}_D$ the Dirichlet projector, the transferred values are the projected current position $\hat{\boldsymbol{x}} = \boldsymbol{P}_D(\boldsymbol{X}_N+\boldsymbol{u}_N)$, velocity $\hat{\boldsymbol{v}} = \boldsymbol{P}_D\boldsymbol{v}_N$, and acceleration $\hat{\boldsymbol{a}} = \boldsymbol{P}_D\boldsymbol{a}_N$ of the partner, relaxed as in Sec. 7. A tied condition prescribes $\boldsymbol{u}_i = \hat{\boldsymbol{x}}_i - \boldsymbol{X}_i$, $\boldsymbol{v}_i = \hat{\boldsymbol{v}}_i$, $\boldsymbol{a}_i = \hat{\boldsymbol{a}}_i$ on every node of the side set, all three components fixed. A frictionless condition prescribes the normal component only,

$$\boldsymbol{x}_i \leftarrow (\boldsymbol{I}-\boldsymbol{n}_i\boldsymbol{n}_i^T)\,\boldsymbol{x}_i+\boldsymbol{n}_i\boldsymbol{n}_i^T\,\hat{\boldsymbol{x}}_i, \qquad \boldsymbol{v}_i \leftarrow (\boldsymbol{I}-\boldsymbol{n}_i\boldsymbol{n}_i^T)\,\boldsymbol{v}_i+\boldsymbol{n}_i\boldsymbol{n}_i^T\,\hat{\boldsymbol{v}}_i, \qquad \boldsymbol{a}_i \leftarrow (\boldsymbol{I}-\boldsymbol{n}_i\boldsymbol{n}_i^T)\,\boldsymbol{a}_i+\boldsymbol{n}_i\boldsymbol{n}_i^T\,\hat{\boldsymbol{a}}_i, \tag{35}$$

with $\boldsymbol{n}_i$ the normal of a facet of the side set containing node $i$, on the current configuration for finite kinematics. The code requires the normal to be along $\pm\boldsymbol{e}_x$ to $10^{-6}$, fixes the $x$ component, and leaves the $y$ and $z$ components free; a side set whose normal is not along $x$ aborts.

**Neumann side.** With $\boldsymbol{N}_N$ the Neumann projector, the force $\boldsymbol{f}_N = \boldsymbol{N}_N(-\boldsymbol{f}_D^{\text{int}}|_\Gamma)$ of (23) is added to the boundary force of the side set, in full for a tied condition and as its normal component $\boldsymbol{n}_i\boldsymbol{n}_i^T\boldsymbol{f}_{N,i}$ for a frictionless one. The reaction transmitted is the static one, as in the classical exchange.

**Options and restrictions.** `naive stabilized: true` sets the acceleration of every node of every contact side set to zero after each substep while contact is active. The exchange shares the relaxation of the classical exchange and does not use the interface predictor. There is no friction law: the two options are no tangential transfer and a full tie. The contact state is one flag for all pairs of the simulation, and when it is active every node of the Dirichlet side set is constrained, whether or not it is in contact with the partner; there is no nodal active set, gap function, or inequality enforcement. The surfaces must be declared in advance as side-set pairs; there is no global search and no self-contact. The projectors are not rebuilt during the run, so sliding is not tracked by the transfer operators. A restart is rejected with contact. The iteration stops on the displacement update of Sec. 3.3.

# 11 Verification of the matching exchange

## 11.1 Benchmarks

**Cantilever beam.** A linear elastic bar ($E = 6.895\,\mathrm{GPa}$, $\nu = 0.25$, $\rho = 2768\,\mathrm{kg/m^3}$; $254 \times 25.4 \times 12.7\,\mathrm{mm}$), clamped at $x = 0$ and released at rest from the bent shape $u_y = 0.3937\,x^2$ (tip deflection 25.4 mm), with initial energy 1506 J and a first bending period of about 10 ms. It is split at $x = 127\,\mathrm{mm}$ into a clamped part and a free part. The free part is meshed with HEX8 elements of 6.35 mm; the clamped part at 6.35 mm (conforming), 8.47 mm (ratio 1:0.75, facets not nested), or 12.7 mm (ratio 1:0.5, nested). Stops of $1\,\mu$s (or $0.5\,\mu$s where stated), 10 ms. The first letter of an integrator pair is the clamped part.

**Nested cylinders.** A steel shell (outer radius 88.9 mm, height 330 mm), a soft filler of one fiftieth of the shell's wave impedance, and an offset core, all neohookean, launched at $-35\,\mathrm{m/s}$ against a support that fixes the bottom face in $z$. The interface is a closed surface inside the filler around the core (a lateral surface and two caps); the inner mesh is 5.1 mm and the outer 6.35 mm (ratio 1:0.8, not nested), 17 to 26 thousand elements per subdomain. Central difference on both sides at a Courant number of 0.5, stops of $0.5\,\mu$s, 2 ms. A linear elastic version with the same meshes (filler $E = 0.214\,\mathrm{GPa}$, $\nu = 0.494$) moves at $-5\,\mathrm{m/s}$ for 1 ms. The inner side is the Dirichlet side.

**Ground truth.** The undecomposed beam keeps $E$ and $\tilde{E}$ within $1.7 \times 10^{-13}$ over 10 ms with the trapezoidal rule and within $2.7 \times 10^{-14}$ with central difference; the trapezoidal value requires a direct linear solver and a small tolerance on the initial-acceleration solve.

## 11.2 Equal steps

Figure 1 shows the energy of the conforming beam for the classical and the matching exchange. Table 1 gives the largest changes of $E$ and $\tilde{E}$ over 10 ms at a Schwarz tolerance of $10^{-12}$ with the coupled initial acceleration.

Table 1: Matching exchange on the beam at equal steps ($1\,\mu$s, 10 ms, tolerance $10^{-12}$, fixed relaxation 0.5, coupled initial acceleration, velocity constraint unless marked $u$). $\delta E$ and $\delta\tilde{E}$ are the largest $|E(t)/E(0)-1|$ and $|\tilde{E}(t)/\tilde{E}(0)-1|$; jump is the largest relative one-sided interface displacement jump over the last 100 stops; it. is the mean and maximum number of Schwarz iterations per stop; time is the elapsed time of the run in seconds. D is the Dirichlet side: the free part, or the implicit side of a mixed pair.

| meshes | pair | $\delta E$ | $\delta\tilde{E}$ | jump | it. | time (s) |
|---|---|---|---|---|---|---|
| conforming | II, $u$ | 2.2e−11 | 5.3e−11 | 5.4e−13 | 7.1/8 | 215 |
| conforming | II | 2.8e−11 | 7.0e−11 | 1.7e−12 | 7.0/8 | 206 |
| conforming | EE | 1.2e−14 | 3.6e−14 | 7.5e−15 | 1.0/1 | 46 |
| conforming | IE | 1.2e−3 | 6.3e−12 | 9.6e−4 | 29.0/32 | 417 |
| conforming | EI | 3.7e−3 | 4.2e−10 | 9.3e−4 | 27.1/28 | 419 |
| 1:0.5 | II, $u$ | 1.2e−10 | 4.3e−11 | 9.8e−13 | 17.3/19 | 437 |
| 1:0.5 | II | 1.2e−10 | 1.8e−11 | 5.6e−12 | 17.2/18 | 436 |
| 1:0.5 | EE | 3.5e−11 | 6.9e−11 | 1.7e−11 | 23.7/26 | 169 |
| 1:0.75 | II | 1.7e−11 | 5.2e−11 | 1.5e−12 | 13.5/14 | 350 |
| 1:0.75 | EE | 2.2e−11 | 6.1e−11 | 6.0e−12 | 16.2/17 | 157 |

The matched quantity is constant to between $10^{-14}$ and $4 \times 10^{-10}$ in every case: $\tilde{E}$ for every pair, and $E$ as well for identical integrators. The residual is the tolerance accumulated over $10^4$

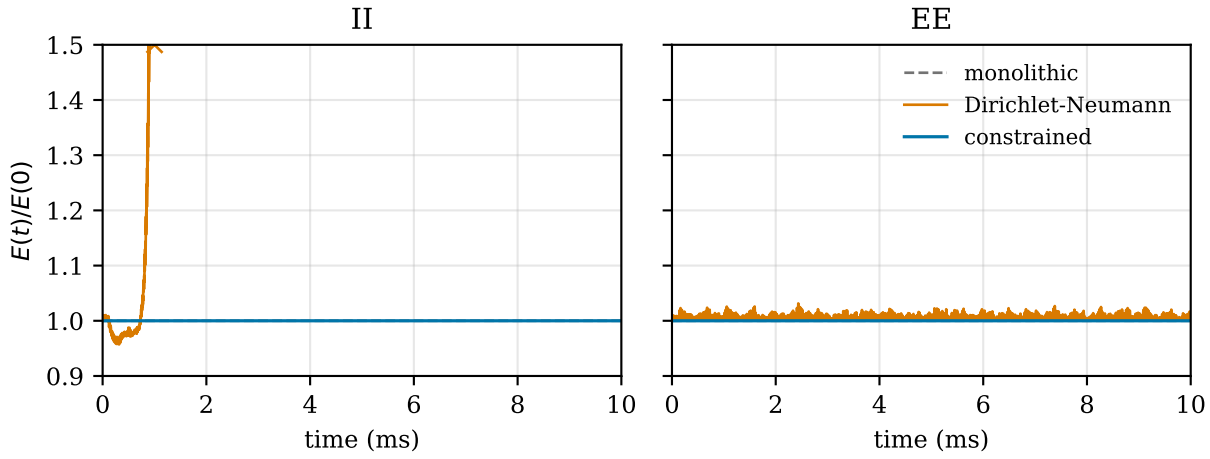



Figure 1: Energy of the conforming beam, stops of 1 $\mu$s, 10 ms: the monolithic reference, the classical Dirichlet–Neumann exchange (Schwarz tolerance $10^{-8}$ on the displacement update, recursive Aitken), and the matching exchange (tolerance $10^{-12}$, Anderson acceleration). The classical exchange inverts an element at 1.0 ms with implicit subdomains (cross); the matching exchange coincides with the reference.

stops; the explicit pair at $\theta = 0.5$ converges in one iteration exactly. For the mixed pairs $E$ varies by 0.1 to 0.4%, which is the expected consequence of the velocity constraint between different integrators, and the root mean square displacement jump stays near $10^{-5}$ m (0.1 to 0.2% of the initial interface displacement), growing by at most 10% over the 10 ms, the displacement drift that velocity continuity permits [7]. At 1:0.75 the projector onto the coarse side reproduces constants only to $1.2 \times 10^{-2}$; the exchange is unaffected because only $\Pi_D$, whose error is $7 \times 10^{-16}$, enters it. The iteration counts of Table 1 were measured with an absolute floor on the root mean square jump equal to the displacement tolerance divided by the step; with the relative criterion alone the conforming II pair takes 8.85 Schwarz iterations per stop (maximum 10) instead of 7.97 (maximum 8), with $\delta E = 7.3 \times 10^{-13}$ and $\delta\tilde{E} = 1.3 \times 10^{-12}$.

Table 2 gives the gain (31) on which the role rules of Sec. 8.7 rest.

Table 2: Eigenvalues of the gain (31) at $\Delta t = 10^{-6}$ s and the spectral radius $\rho$ of the relaxed iteration. Conforming: the meshes of the cantilever example; 1:0.5 and 1:0.75: free part 6.35 mm, clamped part 12.7 and 8.47 mm. D is the Dirichlet side.

| meshes | pair, D | eigenvalues | $\rho(0.5)$ | $\rho(0.75)$ | $\rho(1)$ |
|---|---|---|---|---|---|
| conforming | II, either | $[-1.09, -0.92]$ | 0.05 | 0.57 | 1.09 |
| conforming | EE, either | $-1$ | 0 | 0.50 | 1.00 |
| conforming | IE or EI, D explicit | $[-13.3, -1.63]$ | 6.1 | 9.7 | 13.3 |
| conforming | IE or EI, D implicit | $[-0.61, -0.075]$ | 0.46 | 0.21 | 0.61 |
| 1:0.5 | II, D fine | $[-0.55, 0]$ | 0.50 | 0.25 | 0.55 |
| 1:0.5 | II, D coarse | $[-1.99, -1.83]$ | 0.50 | 1.24 | 1.99 |
| 1:0.5 | EE, D fine | $[-0.50, 0]$ | 0.50 | 0.25 | 0.50 |
| 1:0.5 | EE, D coarse | $[-8.0, -2.0]$ | 3.5 | 5.8 | 8.0 |
| 1:0.75 | II, D fine | $[-0.82, 0]$ | 0.50 | 0.37 | 0.82 |
| 1:0.75 | EE, D fine | $[-0.75, 0]$ | 0.50 | 0.31 | 0.75 |

### 11.3 Different time steps

The beam with the free part subcycled 4:1 (coarse step 1 $\mu$s on the clamped part, fine step 0.25 $\mu$s on the free part) was run with each side as the Dirichlet side, and on the 1:0.5 meshes in two assignments: (A) the clamped part, coarse in mesh and step, as the Dirichlet side; (B) the free part, fine in mesh, given the coarse step and the Dirichlet role, with the clamped part subcycled. Table 3 gives the results at a fixed relaxation of 0.5; Fig. 2 shows the energy of the conforming runs under Anderson acceleration.

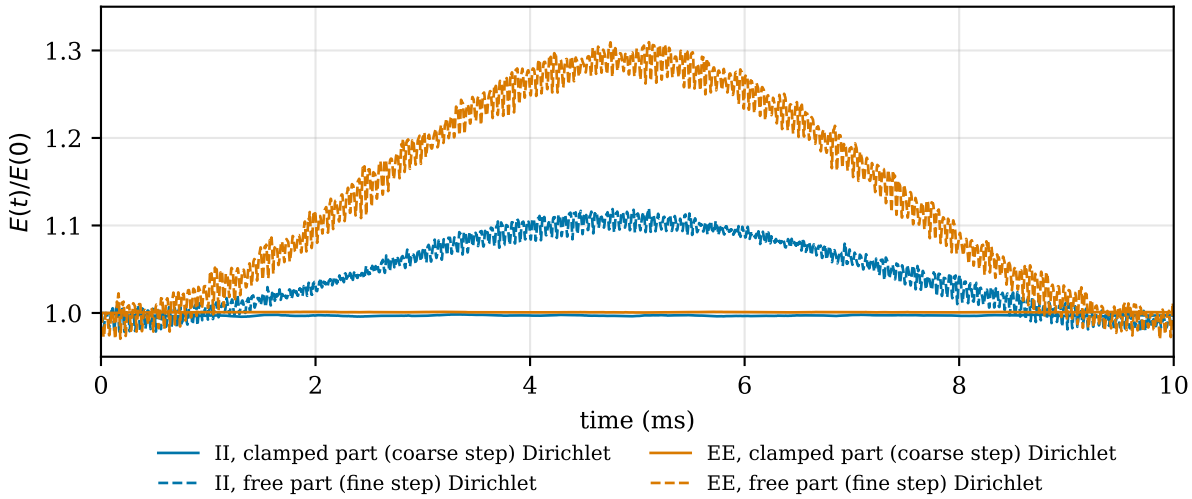



Figure 2: Energy of the conforming beam with the free part subcycled 4:1, tolerance $10^{-12}$, Anderson acceleration. With the clamped part (coarse step) as the Dirichlet side the energy stays within 0.4%; with the free part (fine step) as the Dirichlet side it rises by 12% (II) and 31% (EE) and returns over the first bending period, while $\tilde{E}$ stays constant to $10^{-9}$ in both arrangements.

With the coarse-step side as the Dirichlet side, $\tilde{E}$ is constant to between $4.5 \times 10^{-13}$ and $3.7 \times 10^{-12}$ and the impulse residual (29) is at roundoff. With the fine-step side as the Dirichlet side, $\tilde{E}$ changes by 2 to $4 \times 10^{-9}$ (the accumulated stopping error, which falls with the tolerance), the relative impulse residual has a median of 0.67 (II) and 0.95 (EE), its net value in the bending direction sums to $-17$ and $-29$ N s over the run while the other components stay below $5 \times 10^{-12}$ N s, and $E$ varies by 20 and 51%. Assignment (A) with explicit sides has gain eigenvalues in $[-7.9, -2.0]$; it diverges at every fixed factor from 0.25 up, already in the coupled initial acceleration, and converges with Anderson acceleration (Table 4). Assignment (B) satisfies both role rules and has gain eigenvalues in $[-0.56, 0]$.

### 11.4 Relaxation and acceleration

Table 4 gives the Schwarz iterations per stop under the three relaxation schemes at a tolerance of $10^{-12}$. Aitken acceleration settles at factors near $\theta^*$ of Sec. 8.7: 0.33 on (A) implicit, 0.16 to 0.33 on (A) explicit, 0.64 to 0.84 on (B), and 0.07 to 0.21 with an explicit Dirichlet side. Anderson acceleration at depth 10 needs fewer iterations than Aitken in every case, converges where every fixed factor diverged, and with an explicit Dirichlet side removes the drift of $\tilde{E}$ that Aitken left through stops accepted at the floor of the force residual ($6.8 \times 10^{-11}$ against $6.5 \times 10^{-14}$). The

Table 3: Beam with one side subcycled 4:1, 10 ms, $\theta = 0.5$, relative tolerance $10^{-12}$, no absolute tolerance. D: the Dirichlet side; "coarse" and "fine" refer to the step. $E(t)/E(0)$ at 1, 2, 5, and 10 ms; $\delta E = \max_t |E/E(0) - 1|$ with the sign of the final value; drift: root mean square interface displacement jump at 10 ms (m); impulse: largest net residual per window in $y$ (N s) and the median relative residual; it.: Schwarz iterations per stop, mean/maximum (each iteration advances the fine side four substeps); time: elapsed time of the run in seconds on four threads. The 1:0.5 (A) run with explicit sides ended at the first stop. Lower part: the spectral radius $\rho$ of the relaxed iteration from the linearized gain at a stop.

| meshes | pair, D | 1 | 2 | 5 | 10 | $\delta E$ | $\delta \tilde{E}$ | drift | impulse $y$, rel. | it. | time |
|---|---|---|---|---|---|---|---|---|---|---|---|
| conf. | EE, D coarse | 1.0005 | 1.0009 | 1.0003 | 1.0006 | +1.1e−3 | −4.5e−13 | 3.1e−6 | 1.9e−15, 1.1e−13 | 6.0/7 | 139 |
| conf. | EE, D fine | 1.0432 | 1.1376 | 1.4698 | 0.9938 | −5.1e−1 | +4.4e−9 | 4.2e−6 | 4.8e−2, 0.95 | 36.6/42 | 564 |
| conf. | II, D coarse | 0.9968 | 0.9982 | 0.9972 | 0.9971 | −3.8e−3 | +1.6e−12 | 3.1e−6 | 2.6e−14, 2.2e−13 | 10.1/15 | 560 |
| conf. | II, D fine | 1.0168 | 1.0682 | 1.1977 | 0.9872 | −2.0e−1 | +2.0e−9 | 5.5e−14 | 4.9e−2, 0.67 | 35.7/40 | 1763 |
| 1:0.5 | EE, (A) | diverged in the coupled initial acceleration; ended at the first stop | | | | | | | | | |
| 1:0.5 | EE, (B) | 0.9994 | 0.9995 | 0.9993 | 0.9993 | −8.9e−4 | +3.7e−12 | 1.5e−6 | 2.8e−17, 1.9e−16 | 26.9/31 | 315 |
| 1:0.5 | II, (A) | 0.9989 | 0.9992 | 0.9984 | 0.9993 | −1.9e−3 | −4.7e−13 | 2.2e−6 | 5.9e−14, 4.2e−13 | 44.5/54 | 2555 |
| 1:0.5 | II, (B) | 1.0008 | 1.0009 | 1.0008 | 1.0008 | +1.7e−3 | +1.6e−12 | 1.6e−6 | 2.4e−17, 1.9e−16 | 18.8/21 | 829 |

| meshes | pair | Dirichlet side | $\rho(0.25)$ | $\rho(0.5)$ | $\rho(0.75)$ | $\rho(1)$ |
|---|---|---|---|---|---|---|
| conforming | EE | clamped, coarse step | 0.50 | 0.008 | 0.50 | 1.00 |
| conforming | II | clamped, coarse step | 0.50 | 0.083 | 0.63 | 1.17 |
| 1:0.5 | EE | (A) clamped, coarse mesh and step | 1.23 | 3.47 | 5.70 | 7.93 |
| 1:0.5 | II | (A) clamped, coarse mesh and step | 0.26 | 0.55 | 1.32 | 2.09 |
| 1:0.5 | EE | (B) free, fine mesh, coarse step | 0.75 | 0.50 | 0.25 | 0.50 |
| 1:0.5 | II | (B) free, fine mesh, coarse step | 0.75 | 0.50 | 0.25 | 0.56 |

energies agree across the schemes to the digits measured. A depth of 5 is too short for an explicit Dirichlet side, whose window needs 12 to 16 iterations.

Table 4: Schwarz iterations per stop (mean/maximum) and iterations of the coupled initial acceleration, beam, 10 ms, tolerance $10^{-12}$. Equal steps at 0.5 $\mu$s on conforming meshes; (A) and (B) as in Sec. 11.3 at 1 $\mu$s with 4:1 subcycling; EI and IE with the explicit side as the Dirichlet side at equal steps.

| case | fixed 0.5 | Aitken | Anderson, $m = 5$ | Anderson, $m = 10$ | initial, Aitken | initial, Anderson |
|---|---|---|---|---|---|---|
| equal steps, II | 5.9/6 | 5.8/6 | 5.0/6 | 5.0/6 | 3 | 2 |
| equal steps, EE | 1.0/1 | 1.0/1 | – | – | – | – |
| (A) II | 44.5/54 | 8.8/15 | 6.0/8 | 6.0/8 | 7 | 4 |
| (A) EE | diverged | 20.9/87 | 8.9/9 | 8.9/9 | 21 | 7 |
| (B) II | 18.8/21 | 7.0/9 | 5.0/7 | 5.0/7 | 5 | 3 |
| (B) EE | 26.9/31 | 10.9/13 | 5.0/5 | 5.0/5 | 15 | 4 |
| EI, D explicit | diverged | 34.6/45 | 69.2/118 | 16.0/16 | 47 | 9 |
| IE, D explicit | diverged | 33.9/46 | 53.6/108 | 13.9/14 | 47 | 8 |

### 11.5 Direct interface solve

Table 5 compares the direct solve (32) with the iteration and with the monolithic runs. It conserves $E$ and $\tilde{E}$ to roundoff on the beam at every mesh ratio and on the linear cylinders, in less elapsed time than the iteration on nonconforming meshes; on the cylinders the relative interface velocity jump at the end of every stop is at most $7 \times 10^{-16}$, against $10^{-12}$ for the iteration at $\epsilon = 10^{-12}$, which needs 26.9 Schwarz iterations per stop at a fixed factor of 0.5 and conserves both quantities

to the same digits. On the linear cylinders $\tilde{E}$ varies more than $E$ because the deformation gradient is formed from the current coordinates $\boldsymbol{X} + \boldsymbol{u}$, which loses digits in proportion to $|\boldsymbol{X}|/|\boldsymbol{u}|$; its per-step increments have a standard deviation of $1.1 \times 10^{-13}$ and no significant mean.

Table 5: Direct interface solve, two explicit sides at equal steps. Beam: 10 ms, 1 $\mu$s; cylinders: 1 ms, 0.5 $\mu$s, eight threads (*one thread). $\delta E$ and $\delta\tilde{E}$ as in Table 1; drift is the root mean square interface displacement jump at the end; time is the elapsed time of the run in seconds.

| case | exchange | $\delta E$ | $\delta\tilde{E}$ | drift (m) | it. | time (s) | tolerance |
|---|---|---|---|---|---|---|---|
| beam, conforming, EE | direct | 1.5e−14 | 4.4e−14 | 1.6e−17 | – | 39 | – |
| | iterated | 1.2e−14 | 3.6e−14 | 4.8e−17 | 1.0/1 | 45 | rel. $10^{-12}$ |
| beam, 1:0.5, EE | direct | 7.8e−15 | 4.5e−14 | 1.8e−17 | – | 39 | – |
| | iterated | 2.2e−12 | 4.4e−12 | 4.5e−15 | 26.8/30 | 176 | rel. $10^{-12}$ |
| beam, 1:0.75, EE | direct | 1.2e−14 | 6.0e−14 | 2.7e−17 | – | 39 | – |
| | iterated | 1.6e−12 | 4.6e−12 | 2.3e−15 | 18.1/20 | 164 | rel. $10^{-12}$ |
| linear cylinders, −5 m/s | monolithic | 1.3e−13 | 6.2e−12 | – | – | 371 | – |
| | direct | 9.9e−14 | 3.2e−12 | 6.0e−19 | – | 735 | – |
| | iterated, $\theta = 0.5$ | 1.3e−13 | 1.8e−12 | 8.3e−17 | 26.9/28 | 14 265* | rel. $10^{-12}$ |
| linear cylinders, −35 m/s | monolithic | 1.2e−14 | 4.8e−13 | – | – | 359 | – |
| | direct | 1.1e−14 | 4.3e−13 | 4.3e−18 | – | 731 | – |

## 11.6 Nonlinear material

On the neohookean cylinders the direct solve reduces the relative interface velocity jump to $7\times10^{-16}$ at every step, and $E$ stays within $6.4 \times 10^{-5}$ over 2 ms, against $5.4 \times 10^{-5}$ for the monolithic mesh at the same step. Halving the step reduces both variations by a factor of 3.9 and 4.0 (Table 6, Fig. 3): the variation is the $\Delta t^2$ error of central difference, not interface work. The two signs differ because the two discrete systems have different modified energies.

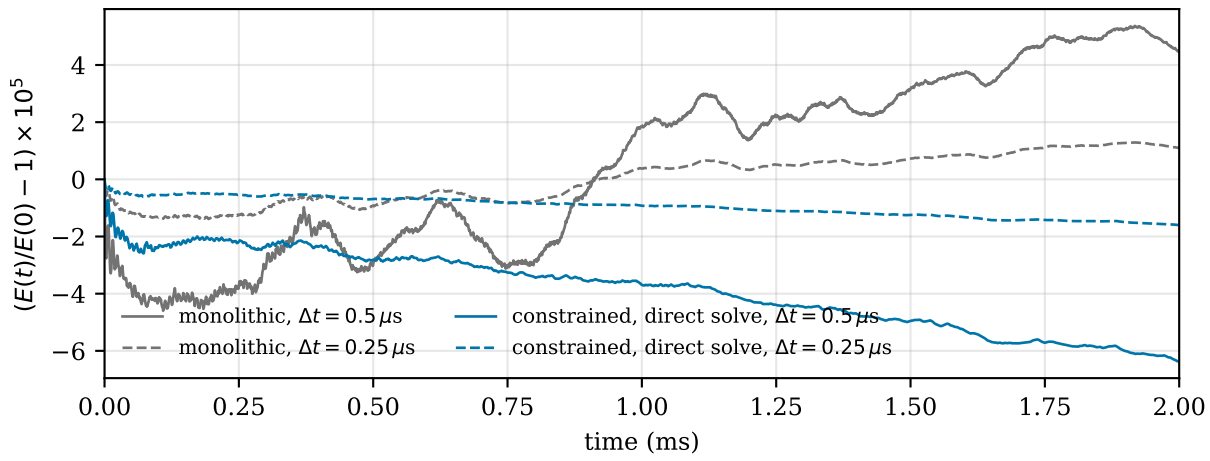



Figure 3: Energy of the neohookean cylinders, −35 m/s, 2 ms: the monolithic mesh and the decomposed mesh with the matching exchange and the direct interface solve, each at two time steps.

Table 6: Neohookean cylinders, $-35$ m/s, central difference, 2 ms: $E/E(0)$ at 0.25, 0.5, 1, and 2 ms and the largest $|E/E(0) - 1|$, for the monolithic mesh and the decomposed mesh with the velocity constraint and the direct interface solve (32), at two steps.

| system | $\Delta t$ (s) | 0.25 ms | 0.5 ms | 1 ms | 2 ms | max $\lvert\delta E\rvert$ |
|---|---|---|---|---|---|---|
| monolithic | $5 \times 10^{-7}$ | 0.999963 | 0.999971 | 1.000019 | 1.000045 | $5.4 \times 10^{-5}$ |
| monolithic | $2.5 \times 10^{-7}$ | 0.999987 | 0.999991 | 1.000004 | 1.000011 | $1.4 \times 10^{-5}$ |
| decomposed, direct | $5 \times 10^{-7}$ | 0.999978 | 0.999972 | 0.999963 | 0.999936 | $6.4 \times 10^{-5}$ |
| decomposed, direct | $2.5 \times 10^{-7}$ | 0.999995 | 0.999993 | 0.999991 | 0.999984 | $1.6 \times 10^{-5}$ |

### 11.7 Inexact subdomain solves

Every Schwarz iteration solves each subdomain to the Newton tolerance of its solver, although only the converged Schwarz iterate is used. Inexact Schwarz solves the subdomains to a large Newton tolerance and relies on the Schwarz iteration, stopped at a small $\epsilon$, for the accuracy. Its effect was measured on the beam of Sec. 11.1 with a neohookean material of the same moduli, so that Newton takes more than one iteration per solve: two trapezoidal subdomains, the matching exchange with Anderson acceleration, $\epsilon = 10^{-12}$, $10^4$ stops of 1 $\mu$s on the mesh of level 1 (180 elements) and $2 \times 10^4$ stops of 0.5 $\mu$s on the mesh of level 2 (1440 elements). The reference run has the relative Newton tolerance $10^{-12}$; the error of a run is $\|u - u_{\text{ref}}\|_2 / \|u_{\text{ref}}\|_2$ over all nodes at 10 ms. Table 7 gives the Newton iterations of each run relative to the reference, the Schwarz iterations per stop, and the error, for constant Newton tolerances, for one Newton iteration per solve carried forward unconverged, and for the tolerance ratio $\eta$ of Sec. 3.5.

Table 7: Neohookean beam, two trapezoidal subdomains, matching exchange, $\epsilon = 10^{-12}$, 10 ms: Newton iterations relative to the reference (Newton tolerance $10^{-12}$), Schwarz iterations per stop, and the relative displacement error against the reference at 10 ms, on the meshes of levels 1 and 2.

| Newton setting | level 1, $10^4$ stops | | | level 2, $2 \times 10^4$ stops | | |
|---|---|---|---|---|---|---|
| | Newton | Schwarz | error | Newton | Schwarz | error |
| tolerance $10^{-10}$ | 0.930 | 7.97 | $1.7 \times 10^{-10}$ | | | |
| tolerance $10^{-6}$ | 0.667 | 7.97 | $6.0 \times 10^{-9}$ | 0.670 | 7.93 | $3.1 \times 10^{-9}$ |
| tolerance $10^{-4}$ | 0.663 | 7.97 | $1.7 \times 10^{-5}$ | 0.640 | 7.93 | $4.8 \times 10^{-5}$ |
| one Newton iteration | 0.334 | 7.97 | $5.3 \times 10^{-3}$ | 0.335 | 7.93 | $3.9 \times 10^{-3}$ |
| tolerance ratio $\eta = 0.01$ | 0.735 | 8.52 | $1.4 \times 10^{-12}$ | 0.717 | 8.27 | $1.8 \times 10^{-12}$ |
| tolerance ratio $\eta = 0.1$ | 0.841 | 10.18 | $3.4 \times 10^{-12}$ | 0.826 | 9.98 | $2.5 \times 10^{-12}$ |

A constant Newton tolerance does not change the Schwarz iteration count, and its error equals the Newton error, not the Schwarz stopping error. The reason is the structure of the iteration: every Schwarz iteration restarts the subdomain solves from the state at the start of the stop, not from the previous Schwarz iterate, so with a fixed Newton setting each Schwarz iteration applies the same map and the iteration converges to the fixed point of that map, which carries the Newton error. The Schwarz criterion measures the change between iterates and does not detect it: with one Newton iteration per solve the iteration meets $\epsilon = 10^{-12}$ in 8 iterations per stop and the result is wrong by 0.5%. The tolerance ratio removes the error at 0.72 to 0.74 of the Newton iterations of the reference. The larger ratio, $\eta = 0.1$, raises the Schwarz count by 26 to 28% and removes most of the saving. The `tight final solve` option, which accepts Schwarz convergence only at an iteration whose Newton tolerance was $10^{-12}$, was triggered at 50 of the $10^4$ stops with $\eta = 0.01$ and did not reduce the error. On a machine with no other load the elapsed time of the level 1 runs

relative to the reference (1081 s on 4 threads) is 0.49 for one Newton iteration per solve, 0.74 for the tolerance $10^{-4}$, and 0.81 for $\eta = 0.01$, within 2.4% of the ratios of residual and tangent assemblies. The energy of the neohookean beam under the trapezoidal rule falls by 5.0% over the 10 ms at both levels, identically in the monolithic run; the loss is divided by 3.6 to 3.9 at each halving of the step on a fixed mesh, so it is the second-order error of the integrator at the Courant number the levels share, and the energy is not a measure of the coupling for this material.

On a static problem the result differs. For a neohookean inclusion in a neohookean matrix (`examples/nonoverlap/static-same-step/matrix-inclusion` at 5% strain in 10 load steps, the classical exchange with Aitken secant relaxation, $\epsilon = 10^{-12}$) Newton tolerances from $10^{-6}$ to $10^{-10}$ save 15% of the Newton iterations at an error of $3\times10^{-13}$, whereas $10^{-4}$ and larger raise the Schwarz count from 36 to 84 or more per load step and reach the maximum of 128, and the tolerance ratio $\eta = 0.1$ saves 24% at an error of $6 \times 10^{-13}$. A warm-started form of inexact Schwarz, in which each subdomain solve starts from its previous Schwarz iterate so that one Newton iteration per Schwarz iteration converges to the solution of the coupled problem, is not implemented; it requires the predictor state of the integrator to be carried across Schwarz iterations.

### 11.8 The classical exchange

The classical Dirichlet–Neumann exchange of Sec. 7 departs from the constrained problem in the five respects listed in Sec. 8: it transmits the static reaction $-\boldsymbol{f}^{\text{int}}$, builds its two transfers independently, imposes displacement, velocity, and acceleration together, stops on the displacement update of the whole domain, and computes the initial acceleration per subdomain with the interface held fixed. The relative difference between the force it applies on the Neumann side and the transposed d'Alembert reaction of the Dirichlet side, measured in the $\boldsymbol{W}_N^{-1}$ norm at the end of every stop, has the median 0.83 (maximum 4.2) with explicit subdomains and 1.47 (maximum 4.6) with implicit ones; under the matching exchange it is below $4\times10^{-16}$. On the beam the classical exchange inverts an element within 1 ms with implicit subdomains on every mesh ratio and with 4:1 steps, and with explicit subdomains the energy varies by 2.6 to 4.3% (Fig. 1). On the neohookean cylinders it stays within 3% over 2 ms with explicit subdomains.

## 12 Limitations and open questions

(i) *Schwarz stopping error.* Every iterated run stops at a tolerance, and the stopping error accumulates: the conserved quantity drifts by about $10^{-10}$ over $10^4$ stops at $\epsilon = 10^{-12}$, and by about $10^{-9}$ with the fine-step side as the Dirichlet side. How the drift scales with $\epsilon$ and with the number of windows, and whether the criterion (30) needs refinement, has not been measured.

(ii) *Warm start of the subdomain solves.* Each Schwarz iteration restarts the subdomain solves from the state at the start of the stop (Sec. 11.7), which bounds the saving of inexact subdomain solves at a fixed accuracy to the fraction of Newton iterations beyond the second. Starting each solve from the previous Schwarz iterate would let one Newton iteration per Schwarz iteration converge to the coupled solution; its effect on the Schwarz count and on the rate invariant identity has not been measured.

(iii) *Mixed pair with an explicit coarse-step side.* The coarse-step side must be the Dirichlet side and an explicit Dirichlet side has a gain down to $-13$; Anderson acceleration converges at equal steps in this configuration but has not been measured with subcycling.

(iv) *Cause of the energy variation with the fine-step side as the Dirichlet side.* The variation coincides with an impulse residual of the order of the impulse; that the residual causes it has not been tested separately.

(v) *Nonlinear materials.* Conservation of $E$ to roundoff needs an energy-momentum integrator and a midpoint velocity constraint.

(vi) *Higher flux order and window solves.* Flux order $r \geq 2$ in the sense of Connors et al. and a direct interface solve over a subcycled window are not needed for conservation; whether they reduce the iteration count or the elapsed time has not been measured.

(vii) *Implicit subdomains on the cylinders* and more than two subdomains have not been run with the matching exchange.

(viii) *A conserving Robin condition.* A Robin condition assembled from the d'Alembert reaction and the one-sided constraint of Sec. 8.1 has the fixed point of the constrained problem and could serve as a solver of it; it has not been implemented.

(ix) *Contact.* The contact coupling has one active flag for the whole simulation, no nodal active set, no friction law, and transfer operators fixed at initialization (Sec. 10).

# A Input keys, tests, and examples

The keys are documented in `docs/src/reference/multidomain.md` of the Norma repository; unknown keys produce a warning with a suggestion and do not abort. The implementation is in `src/schwarz.jl`, `src/simulation.jl`, `src/boundary_conditions.jl`, `src/interpolation.jl`, and `src/overlap_energy.jl`.

**Controller (`type: multi`).** `domains` (required); `initial time`, `final time`, `time step` (required); `minimum iterations`, `maximum iterations` (required); `relative tolerance`, `absolute tolerance` (required unless every coupling is a matching pair, in which case they are ignored with a warning); `matching relative tolerance` (required when a pair sets `matching`), `matching absolute tolerance` (default 0); `relaxation` (`aitken recursive`, `aitken secant`, `anderson`; absent means fixed), `relaxation parameter` (default 1, or 0.5 under `anderson`), `aitken N0 parameter` (default 1), `anderson depth` (default 10); `interface predictor` (default false); `stalled interface jump action` (`warn`, `abort`); `unconverged step action` (`abort`, `warn`); `inexact Newton` block with `tolerance ratio` (0.01), `minimum tolerance` ($10^{-12}$), `maximum tolerance` ($10^{-2}$), `tight final solve` (false); `naive stabilized` (default false, contact only); `blended energy output` (default false); `Exodus output interval` (default the controller step), `CSV output interval` (default off); `swaps`; `restart`. The keys `constrained`, `constraint`, `constraint absolute tolerance`, and `forcing factor` are retired and abort with a message naming their replacements.

`Schwarz overlap`. `side set`, `source`, `source block` (required); `search tolerance` (default $10^{-6}$); `weak` (default false); `compute overlap L2 relative error` (`disp`, `velo`, `acce`).

`Schwarz DN nonoverlap`. `side set`, `source`, `source side set` (required); `default BC type` (`Dirichlet` or `Neumann`, opposite on the two sides); `swap BC types` (default false); `matching` (`velocity` or `displacement`, the same on both sides; absent selects the classical exchange); `interface solve` (`iterative` or `direct`, the same on both sides).

`Schwarz RR nonoverlap.` `side set`, `source`, `source side set`, `robin parameter` (required, positive).

`Schwarz contact.` `side set`, `source`, `source side set`, `friction type` (`frictionless` or `tied`) (required); `swap BC types` (default false).

**Tests and examples.** Test numbers refer to `test/runtests.jl`. Overlapping coupling: tests 15 (static cuboids), 56 (dynamic cantilever, weak form), 57 and 75 (swaps), 109 (blended energy), 113 (restart), and 117 (quasi-static); examples under `examples/overlap` and `examples/ahead/overlap`. Classical Dirichlet–Neumann exchange: tests 16, 18, 55, 58, 91, 92, 123 (displacement transfer on a curved interface), and 137 (windowed stops); examples `examples/nonoverlap/static-same-step/cuboids-dirichlet-neumann`, `matrix-inclusion`, and `examples/nonoverlap/dynamic-same-step/cantilever-dn`. Matching exchange: tests 135 (rate invariant) and 136 (operators, reaction, Newmark relations, rate invariant conservation, displacement matching, coupled initial acceleration, subcycling, the direct solve, Aitken and Anderson acceleration, the equivalence of Anderson acceleration with GMRES on a probed interface map, and the retired keys); examples `examples/nonoverlap/dynamic-same-step/cantilever-dn-nonconforming` and the linear cylinders of `examples/jmp/concentric-cylinders-linear`; the parameter study `studies/dynamic-stability` measures the effect of each factor (exchange, mesh ratio, tolerance, time steps, solver, refinement) on the beam. Robin–Robin: tests 17, 18, 93, 95, and 137; examples `cuboids-robin-robin` (static and dynamic) and `cantilever-rr`. Contact: tests 19 (transfer operators), 20 (static cubes, frictionless and tied), and 21 (dynamic cubes, tied); examples under `examples/contact`.

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